

Video Casting Study

2020

ANRC

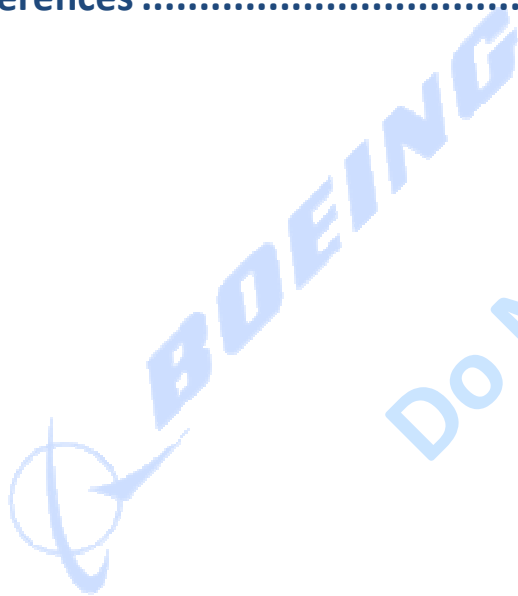
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Introduction

Passengers bring media content on personal portable devices onto airplane cabin, and it is desirable to cast onto seat back displays. Considering that 2.4 & 5 GHz Wi-Fi on the airplane is congested, it is desirable to leverage alternate technology that is/will be supported in commercial, off the shelf portable devices.

Problem Statement: To develop a wireless casting solution that supports high resolution (4K) video bandwidth and scales to 400+ passenger devices in close proximity.

Assumption: The content to be cast is available offline on the user's device

It was necessary to survey the possible methods of casting and their functionality. This will be helpful in defining the plan of action to implement the best methods that will be relevant in upcoming years.

This report contains information on:

- How wireless casting works
- Comparison of various Wi-Fi standards
- Private LTE
- Wired Casting
- Current Hardware trends in commercial casting products and in mobile phones
- Components involved to build a display and its associated cost
- Possible plan of action

Wireless Casting

How wireless casting works:

Wireless casting is split in two planes, the control and the data plane.

The control plane uses the TCP port 4172. Session capabilities like frame rate, media control and also the port for UDP packets are conveyed by the source to the display(sink). RTSP protocols are used to control the media stream by sending instructions for play, pause and navigation.

The data plane is responsible for transferring the media content using UDP and RTP headers. The source runs a codec and recreates the audio and video frames which are merged and packetized as RTP frames as shown below. The sink has an RTP interpreter and a demux which extract the media content. This content is synchronized and played by a media player.

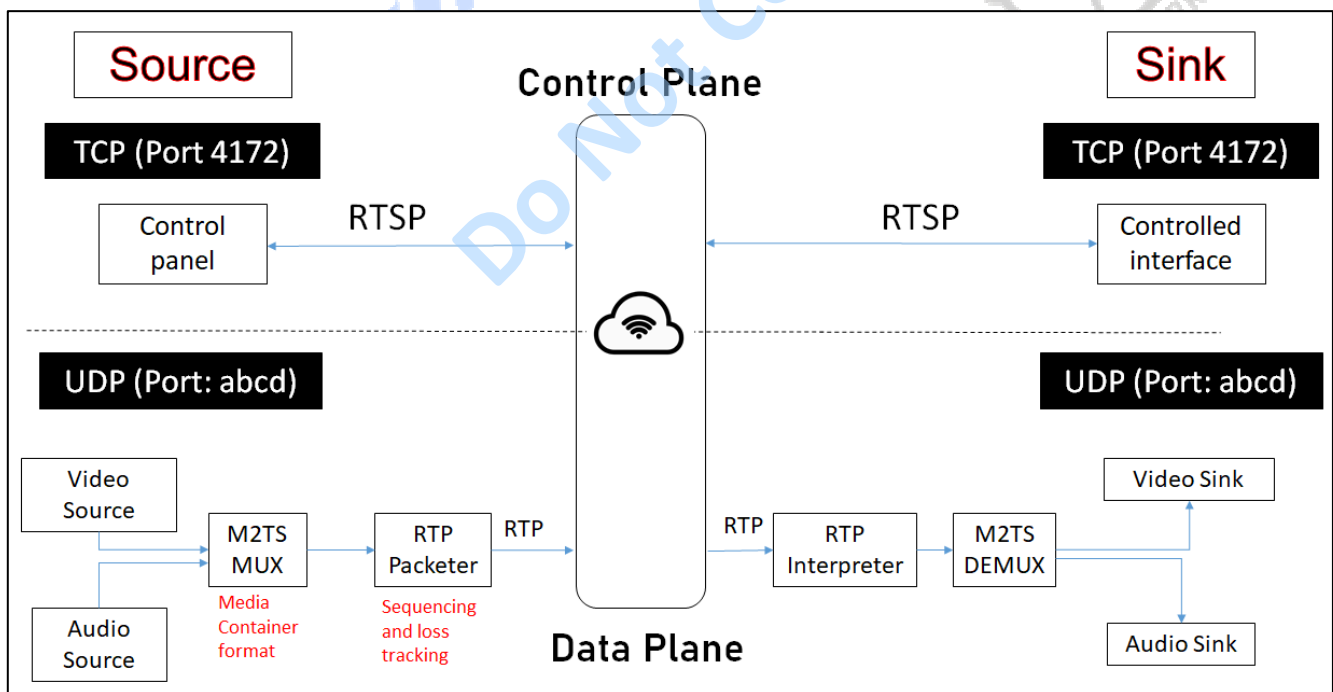


Figure 1 Overview of the Control and Data planes

Wi-Fi Standards (IEEE 802.11 xx)

Specifications		IEEE 802.11ax				IEEE 802.11ad	IEEE 802.11ay
Frequency (GHz)		2.4/5/6				60	60
Bandwidth (MHz)		20	40	80	80+80	2160	8000
Stream data rate (Mbits/s)		Up to 1147	Up to 2294	Up to 4803	Up to 10530	Up to 8000	Up to 20000
MIMO		8				NA	4
Modulation		MIMO-OFDM				OFDM, single carrier, low-power single carrier	OFDM, single carrier
Approx. Range	Indoor	30 m				3.3 m	10m
	Outdoor	120 m				..	100m
Beam forming		Supports beam forming with explicit feedback from client device to ap				Supports directional beam forming	Supports beamforming protocols like SU-MIMO/MU-MIMO

Table 1 IEEE 802.11xx Standards

The nominal channel bandwidth is 2.16 GHz. The useful ISM band around 60 GHz (57 GHz to 66 GHz) is regulated differently in various regions of the world. Six channels are defined for this band, but they are not universally available. Channel 2 is available in all regions and is therefore used as the default channel.

Channels	
1	58.32 GHz
2	60.48 GHz
3	62.64 GHz
4	64.80 GHz
5	66.96 GHz
6	69.12 GHz

Table 2 Channel numbers in the 60 GHz band

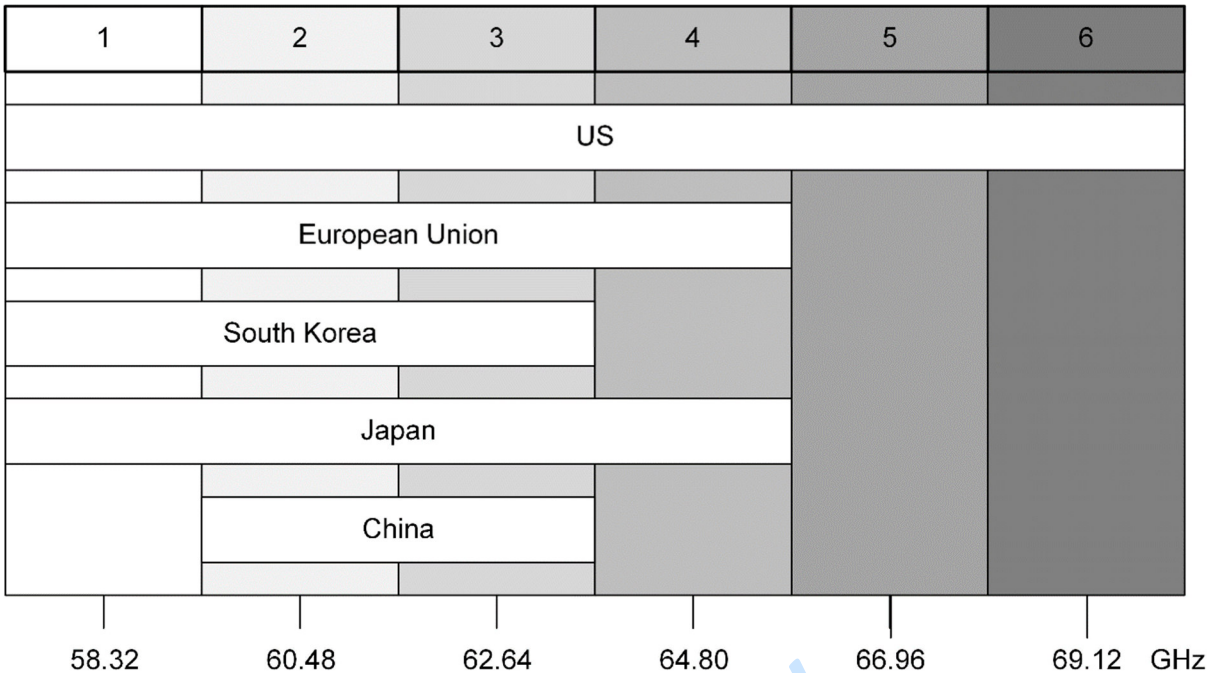


Figure 2 60GHz - Channel Allocation

Observations

802.11ay will not be a new type of WLAN in the IEEE 802.11 set, but will simply be an improvement on IEEE 802.11ad. Where 802.11ad uses a maximum of 2.16 GHz bandwidth, 802.11ay bonds four of those channels together for a maximum bandwidth of 8.64 GHz. MIMO is also added with a maximum of 4 streams. The link-rate per stream is 44Gbit/s, with four streams this goes up to 176Gbit/s.

Note: 802.11ay enabled hardware is not commercially available yet.

Advantages & Disadvantages of using 802.11ad and 802.11ay wireless protocols for video casting

Advantages

- Suited for streaming high quality video
- Bulk data transfers
- It can support data rates up to 1000 Mbps
- No compression required to transfer high quality videos
- Can only function at 30 feet (≤ 10 m)
- Not able to pass through physical obstructions
- Less interference with nearby 60 GHz networks, and makes remote eavesdropping and network security break-ins much difficult
- Government regulatory agencies manage 60 GHz usage around the world but generally do not require devices to be licensed
- Being an unlicensed spectrum, 60 GHz represents a cost and time-to-market advantage for equipment makers that in turn benefits consumers.
- Close to zero latency

Disadvantages

- Radios consume high power
- Devices require line-of-sight between one another for optimal performance
- Very few devices support this currently

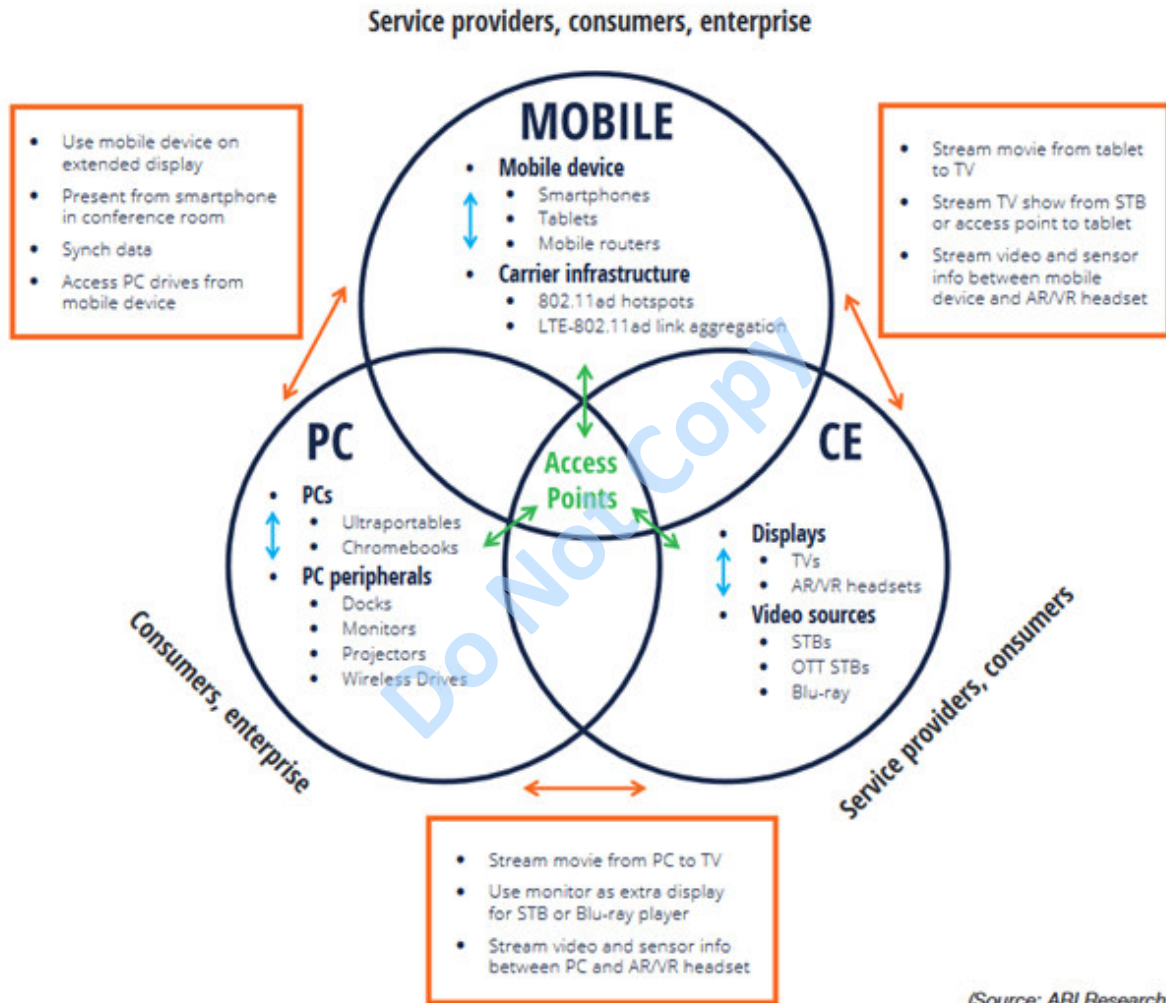
802.11ad (WiGiG)

Reasons Miracast can be extended over 802.11ad to Stream videos in limited space environment

- Video requires steady reliable data rates but can be disrupted by interference and network congestion issues. Existing Wi-Fi spectrum bands are getting congested in homes where many devices and other products are using the Wi-Fi network, including to stream video. Using Miracast over 802.11ad would avoid this congestion.
- Shifting as much video traffic as possible to 802.11ad will offload a considerable amount of traffic from 2.4 GHz and 5 GHz based Wi-Fi networks, which will result in an improved user experience over all three spectrum bands.

- **Fast Session Transfer (FST)** is a MAC layer feature of 802.11ad that allows for multiband operation. While 802.11ad is incompatible with older standards at the PHY layer due to operation in 60 GHz, interoperability is possible at the MAC layer. With FST, we can have seamless transfer of sessions from 60 GHz band to other bands, and vice versa.

802.11ad Product Ecosystem & Use Cases



(Source: ABI Research)

Figure 3 802.11ad Product Ecosystem & Use Cases



Figure 4 Semiconductor Vendors That Offer or Plan to Offer 802.11ad Chipsets

Private LTE

Frequency band: 3.5 GHz

Private LTE is based on technologies developed for wide area mobility and scaled down to be deployable by private organization. This background gives LTE some advantages over WI-FI. These advantages include:

- **Range budget**: As LTE is a cellular technology; it has been designed to operate under fading channels, providing good cell edge performance. As a rule, one LTE cell covers about the same area as two or three Wi-Fi access points.
- **Spectral efficiency**: LTE is designed for mobility and operation over a wide area. It can handle larger delay spreads, which also contribute to overall spectral efficiency
- **Mobility & Security**
- **High to Low Rate Scaling**: From gigabit service to IOT devices for low power
- Capacities can be significantly higher than those of lower frequency bands

A Private LTE network leverages micro towers and small cells — similar to a Wi-Fi access point — on-site to replicate the larger public network. Private LTE can be based on licensed, unlicensed, or shared spectrum.

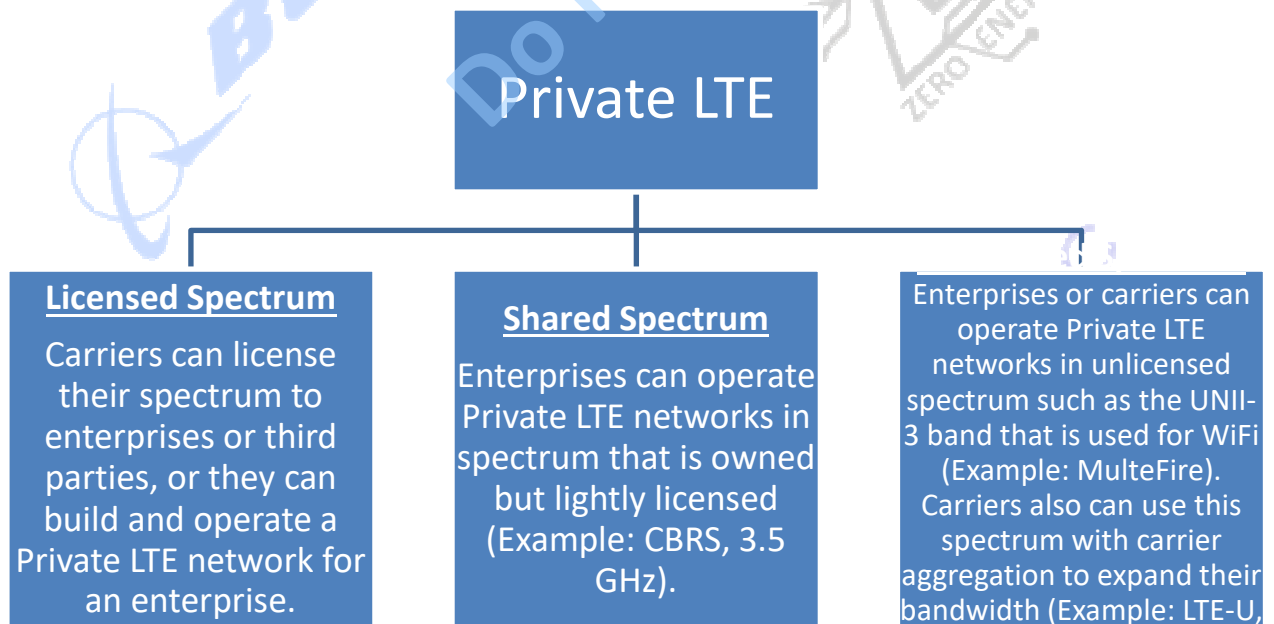


Figure 5 Private LTE Spectrum

Casting Method

DLNA

Digital Living Network Alliance (DLNA) was founded by a group of PC and consumer electronics companies in June 2003 to develop and promote a set of interoperability guidelines for sharing digital media among devices. Media management, device discovery and control are included with the standard of Universal Plug and Play (UPnP).

There are several DLNA certified class of devices

- Digital Media Server (DMS): store content and make it available to networked digital media players (DMP) and digital media renderers (DMR).
- Digital Media Player (DMP): find content on digital media servers (DMS) and provide playback and rendering capabilities.
- Digital Media Renderer (DMR): play content as instructed by a digital media controller (DMC), which will find content from a digital media server (DMS)
- Digital Media Controller (DMC): find content on digital media servers (DMS) and instruct digital media renderers (DMR) to play the content. Content does not stream from or through the DMC.
- Generally, digital media players (DMP) and digital media controllers (DMC) with print capability can print to DMP.

There are also classes of devices similar to the above specific to Mobile devices.

Underlying Stack

Content Protection - DRM

Media transport - Http/RTP/RTSP

Device Discovery & Control - UPnP

Connectivity - Ethernet/802.11/BLE/Wired

DLNA System options

In Pull Type design, DMP pulls the content from the DMS.

Example - Desktop player pulls the content from the DMS server hosted elsewhere.

In Push Type design, DMS & DMC push content to the DMR and provides control methods over content. Example - Handled devices connected to DMS and can view the content.

Casting Methodology comparisons

Features	Miracast and similar technology	DLNA	Chromecast & similar technology
Casting Method	Screen Mirroring over Wi-Fi Direct. Miracast is the wireless version of HDMI	Multimedia Streaming using DLNA & UPnP.	Multimedia Streaming of online content and Screen Mirroring
Device Infra	Android devices, Desktops, Tablets that supports Miracast. Apple devices comes with similar technology Airplay.	Device that stores content and distributes content has to support Digital Media Server (DMS). Devices that plays and renders requires DMP or DMR	Chromecast streaming device
Observation	Streaming apps like Netflix, Prime, YouTube Movies offline contents doesn't cast through screen mirroring because of DRM. Screen Mirroring will drain the mobile device battery relatively sooner compared to streaming.	For mobile devices to stream the offline content requires Media server implemented to stream the content to the Player/Renderer. This requires custom app implemented with DLNA DMS support.	Chromecast primary function is to stream the online content to the Chromecast devices. However, Chromecast offline casting/mirroring of content from mobile device to Chromecast device is to be further explored.

Table 3 Comparison of Casting Methodologies

WebRTC

WebRTC is an emerging HTML5 standard which stands for Web Real Time Communications. WebRTC allows for peer to peer communication between browsers to exchange video, audio and data. WebRTC designed for personal communication application for the internet using only web browsers without any plugins. Until recent, RTC has been dedicated to corporate and complex implement, requires expensive audio and video technologies to be licensed or developed in house. Integrating RTC technology with existing content, data and services has been difficult and time consuming, particularly on the web.

As Gmail video chat became popular in 2008 and in 2011 Google introduced Hangouts, which use the Google Talk service. Google has to acquire GIPS, a company which had developed many components required for RTC, such as codecs and echo cancellation techniques and so on. Google open sourced the technologies developed by GIPS and engaged with relevant standards bodies at the IETF and W3C to ensure industry consensus. In May 2011, Ericsson built the first application of WebRTC.

Need for WebRTC

WebRTC is an open standard for real-time, plugin-free video, audio and data communication. The need was real:

- Many web services used RTC, but needed downloads, native apps or plugins. Those included Skype, Facebook and Google Hangouts.
- Downloading, installing and updating plugins is complex, error prone and annoying.
- Plugins are difficult to deploy, debug, troubleshoot, test and maintain—and may require licensing and integration with complex, expensive technology. It's often difficult to persuade people to install plugins.
- Inter operation between applications developed for desktop and native mobile application is really hard and need telecom developer experience.
- Security concerns to control the use of sensitive and unreliable networks.

Thus, to address these issues WebRTC a great choice for corporate applications. The guiding principles of the WebRTC project is its APIs should be open source, free, standardized, built into web browsers and more efficient than existing technologies.

WebRTC - Pros and Cons

Advantages of WebRTC is users don't need to install any plugins or download any software to stream video and audio directly in the browser. First, WebRTC call (handshake) is carried out between web application and the participants of a WebRTC call. This process is called signaling. After signaling is complete and a web RTC connection has been established between end devices completely peer to peer. This is especially good for data intensive applications like video chat or file sharing, because communication happens directly between the two users independent of platform. No intermediary server is needed to host the data or stream it to other parties. This makes WebRTC a very efficient solution for data or video exchange between two parties. This also means that WebRTC can scale extremely well for small conversation. The load on media servers for signaling is very small. Once the peer to peer connection has established between two party's media server is out of the loop and theoretically build a WebRTC based service that allows for thousands of simultaneous video streams between small groups of people.

In WebRTC, privacy during communication is enabled. Once the peer to peer connection is established between two web RTC clients, all traffic between those two clients is completely encrypted and It is harder to do a middle attack. But it is important to note that the signaling or handshaking required to start a web or RC connection is not encrypted. The signaling process could allow hackers to determine who is in conversations at what times. WebRTC is a great choice for those type of real time communications.

The other important disclaimer needs to be given about WebRTC is that native WebRTC are support is available only in latest version of Chrome, Firefox and Opera. Internet Explorer and Safari do not support web RTC. However, it is expected that Microsoft will release a browser supporting WebRTC. Also, applications based on WebRTC can run in IOS and plugins can be still installed for internet explorer and safari browsers. Some of these negatives will hopefully go away over time.

Applications and Use Case of WebRTC

- **Video Conference:** Building a simple video conferencing tool is a great way to get started with WebRTC and there are still niche markets that can be tackled which could use a custom videoconferencing tool that can help in marketing.
- **Customer Support Centre:** WebRTC may not require any telecommunications developer expertise; but having such expertise WebRTC can use telephone network and create a unified communication solution that combines traditional telephones with Web based communications applications and allows to have a seamless customer support.
- **Medical Support:** WebRTC can be used to make medical applications a huge opportunity in many countries. Around the world telemedicine innovation can be stifled by privacy concerns, security concerns and government regulations. WebRTC has inherent advantages because a WebRTC video chat is over an encrypted peer to peer connection. It is much harder to snoop on private conversation or steal patient data with WebRTC. Also, WebRTC can build a medical application that allows to transfer video and audio data in real time in secured and encrypted fashion.
- **Insurance Claim:** WebRTC can completely revolutionize an Insurance Industry. Car insurance claims are one area that could be improved after an accident occurs. Many insurance companies require an insurance claims agent to visit the scene of the accident before a claim can be filed. Sometimes this even means the accident continues to block traffic until the claims-agent has arrived. Instead, to report insurance company, can call to mobile phone of victim and use mobile camera to capture live video feed and can record footage about the location and conditions of the accident along with the damage to car.
- **Gaming Applications:** Peer to peer data channels of WebRTC allow data exchange in peer to peer manner once signaling is completed. Which means real time data can be communicated over mesh networks without use of complicated media servers. Also, chat application with WebRTC allows to exchange text-based chats in real time communication between the players.
- **File Transfer:** Apart from Video and Audio, WebRTC can allow transfer sensitive files or data directly between two peers without anyone being able to see them. With no server in middle where those files might get hacked. This concept has

been used to build not only file sharing applications directly in the browser but also complete content delivery networks.

WebRTC APIs

APIs of WebRTC technology can be broadly classified into Media capturing APIs and Peer-to-Peer connectivity APIs as shown in the figure.

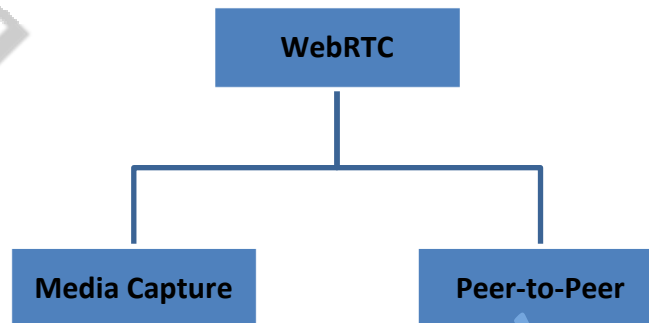


Figure 6 Broad API classification of WebRTC

Media Capture Device

- Media capture devices includes video cameras and microphones, also screen capturing "devices".
- For cameras and microphones, `navigator.mediaDevices.getUserMedia()` to capture Media Streams.
- For screen recording, we use `navigator.mediaDevices.getDisplayMedia()`.

The Peer-to-Peer Connectivity

- The peer-to-peer connectivity is handled by the `RTCPeerConnection` interface
- This is to establishing and controlling the connection between two peers in WebRTC.

Introduction to WebRTC protocols

Interactive Connectivity Establishment

- Interactive Connectivity Establishment (ICE) is a framework to allow your web browser to connect with peers.
- There are many reasons why a straight up connection from Peer A to Peer B simply won't work because it needs to bypass firewalls, NATing, exchange IP address using STUN and TURN servers

Session Traversal Utilities for NAT (STUN)

- Session Traversal Utilities for NAT (STUN) (acronym within an acronym) is a protocol to discover your public address and determine any restrictions in your router that would prevent a direct connection with a peer.
The client will send a request to a STUN server on the Internet who will reply with the client's public address and whether or not the client is accessible behind the router's NAT.

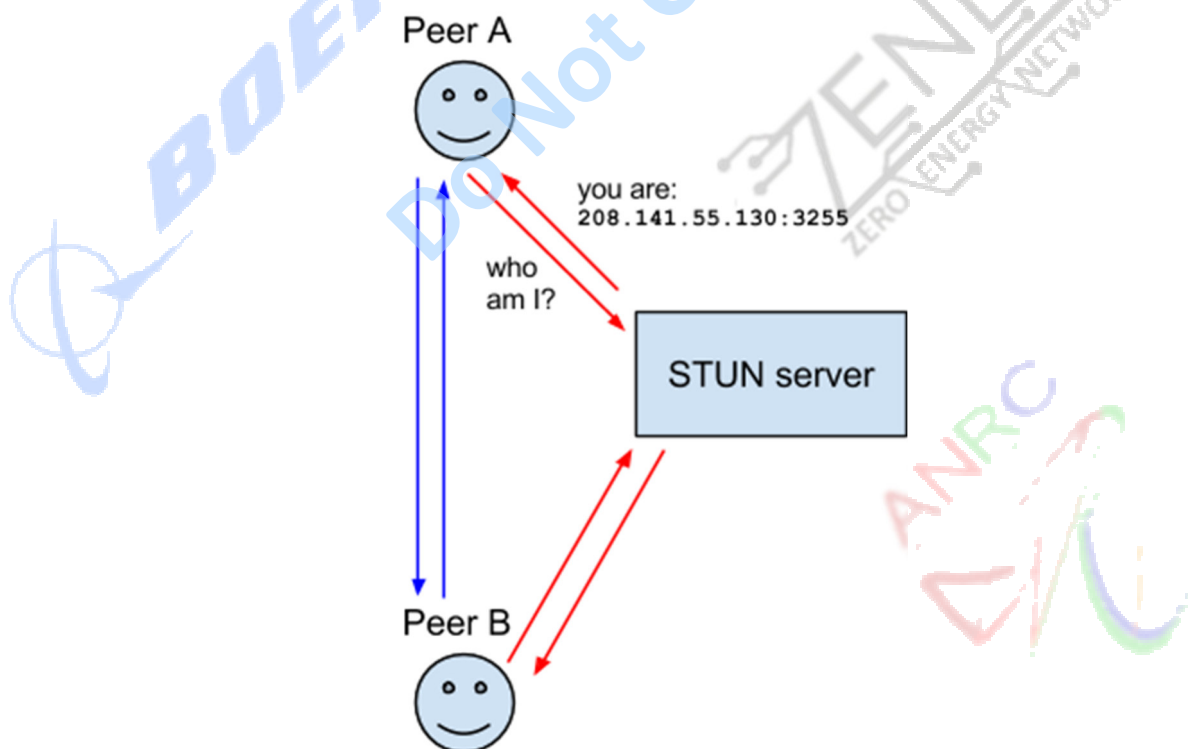


Figure 7 Session Traversal Utilities for NAT (STUN)

Session Description Protocol

Session Description Protocol (SDP) is a standard for describing the multimedia content of the connection such as resolution, formats, codecs, encryption, etc. so that both peers can understand each other once the data is transferring. This is, in essence, the metadata describing the content and not the media content itself.

Signalling in WebRTC

- WebRTC can't create connections without some sort of server in the middle. We call this the signal channel or signalling service. It's any sort of channel of communication to exchange information before setting up a connection, whether by email, post card or a carrier pigeon.
- The information we need to exchange is the Offer and Answer which just contains the SDP.
- Peer A who will be the initiator of the connection, will create an Offer. They will then send this offer to Peer B using the chosen signal channel.
- Peer B will receive the Offer from the signal channel and create an Answer. They will then send this back to Peer A along the signal channel.

ICE candidates

- As well as exchanging information about the media (discussed above in Offer/Answer and SDP), peers must exchange information about the network connection.
- This is known as an ICE candidate and details the available methods the peer is able to communicate (directly or through a TURN server).
- Typically, each peer will propose its best candidates first, making their way down the line toward their worse candidates.
- Ideally, candidates are UDP (since it's faster, and media streams are able to recover from interruptions relatively easily), but the ICE standard does allow TCP candidates as well.

Creating a data channel

- The underlying data transport used by the RTCDataChannel can be created in one of two ways:
 - Create the transport and announce it to the remote peer
 - Write code to negotiate the data transport and write code to signal to the other peer that it needs to connect to the new channel.

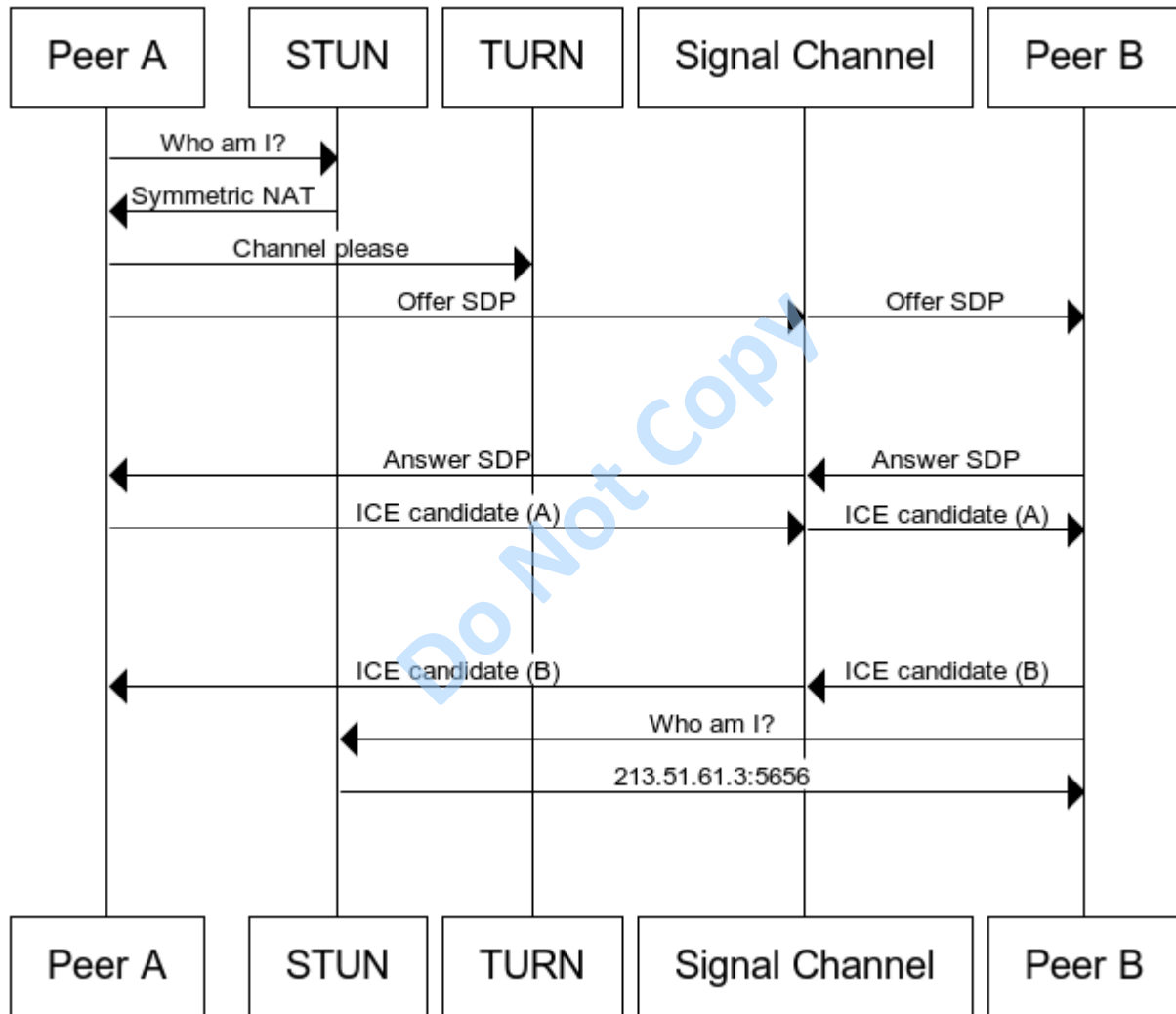


Figure 8 Creating a data channel in WebRTC

Proposed Video Casting Architecture using WebRTC

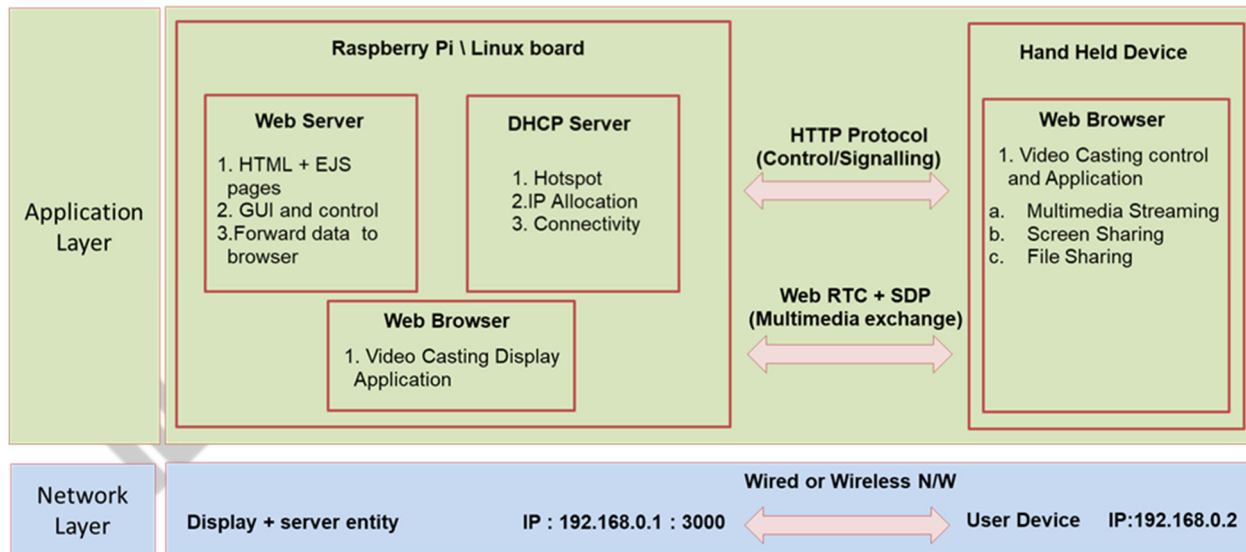


Figure 9 Video Casting using WebRTC - Architecture

Video casting application

- Display + Server entity
- Hand Held Device

Display + Server entity

- **Web server:** Provides Video Casting Application GUI developed using HTML5+EJS on request. Server and Client pages are served
- **DHCP sever:** Facilities the network connectivity i.e. Hotspot, allocating IP, establish connectivity
- **Web Browser:** Web Application to display video casting data

Hand Held device

- **Web Application to control:** Web Application to control video casting services

Video casting network layer and lower layers

- IPv4 or 6 network layers
- Media can be Wired or Wireless

Streaming Protocols

RTSP / RTP

The **Real Time Streaming Protocol** (RTSP) is a network control protocol designed for use in entertainment and communications systems to control streaming media servers. The protocol is used for establishing and controlling media sessions between end points. Clients of media servers issue VHS-style commands, such as play, record and pause, to facilitate real-time control of the media streaming from the server to a client (Video On Demand) or from a client to the server (Voice Recording).

The transmission of streaming data itself is not a task of RTSP. Most RTSP servers use the **Real-time Transport Protocol** (RTP) in conjunction with Real-time Control Protocol (RTCP) for media stream delivery. However, some vendors implement proprietary transport protocols. While similar in some ways to HTTP, RTSP defines control sequences useful in controlling multimedia playback. While HTTP is stateless, RTSP has state; an identifier is used when needed to track concurrent sessions. Like HTTP, RTSP uses TCP to maintain an end-to-end connection and, while most RTSP control messages are sent by the client to the server, some commands travel in the other direction (i.e. from server to client).

- **Audio Codecs:** AAC, AAC-LC, HE-AAC+ v1 & v2, MP3, Speex, Opus, Vorbis
- **Video Codecs:** H.265 (preview), H.264, VP9, VP8
- **Playback Compatibility:** Not widely supported (QuickTime Player and other RTSP/RTP-compliant players, VideoLAN VLC media player, 3Gpp-compatible mobile devices)
- **Benefits:** Low-latency and requires no buffering
- **Drawbacks:** Not optimized for quality of experience and scalability
- **Latency:** 2 seconds
- **Variant Formats:** The entire stack of RTP, RTCP (Real-Time Control Protocol), and RTSP is often referred to as RTSP



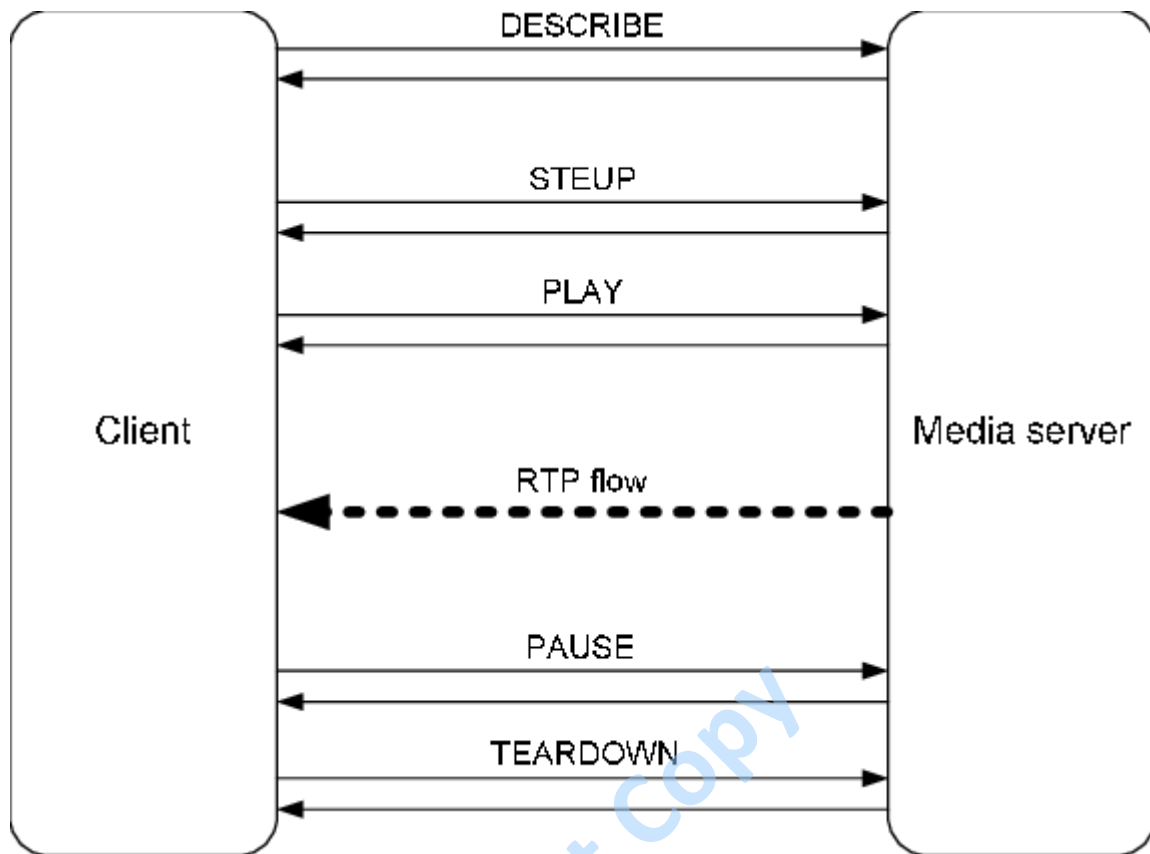


Figure 10 RTSP between Client and Media Server

Testing RTSP/RTP using Wi-Fi

A prototype was developed by making use of open source RTSP server and client code. Using Android simulator, a mobile device environment was created and RTSP server was executed on it. A MJPEG video was stored in the mobile device. RTSP client was also executed in a separate instance.

Once RTSP server came up, control options would be provided to user. When SETUP action was performed it established connection between RTSP server with the client. Once the connection was established, clicking PLAY, played the video that was stored in the mobile device. The current demo would only play MJPEG video formats. The demo video can be accessed using the following link:

<https://youtu.be/diqQeF9mSG8>

RTMP

Real-Time Messaging Protocol (RTMP) was initially a proprietary protocol developed by Macromedia for streaming audio, video and data over the Internet, between a Flash player and a server. Macromedia is now owned by Adobe, which has released an incomplete version of the specification of the protocol for public use. RTMP is a TCP-based protocol which maintains persistent connections and allows low-latency communication.

RTMP maintains a constant connection between the player client and server, allowing the protocol to act as a pipe and rapidly move video data through to the viewer. RTMP sits on top of the Transmission Control Protocol (TCP), it uses a three-way handshake when transporting data. The initiator (client) asks the acceptor (server) to start a connection; the acceptor responds, then the initiator acknowledges the response and maintains a session between either end. For this reason, RTMP is quite reliable.

- **Audio Codecs:** AAC, AAC-LC, HE-AAC+ v1 & v2, MP3, Speex, Opus, Vorbis
- **Video Codecs:** H.264, VP8, VP6, Sorenson Spark®, Screen Video v1 & v2
- **Playback Compatibility:** Not widely supported
 - Limited to Flash Player, Adobe AIR, RTMP-compatible players
 - No longer accepted by iOS, Android, most browsers, and most embeddable players
- **Benefits:** Low-latency and minimal buffering
- **Drawbacks:**
 - Not optimized for quality of experience or scalability
 - While RTMP works well, it has historically encountered issues getting past firewalls.
 - As a stateful protocol, RTMP requires a dedicated streaming server.
 - With Adobe announcing death of Flash the industry now favors HTTP based streaming protocols
- **Latency:** 5 seconds
- **Variant Formats:** RTMPT (tunneled through HTTP), RTMPE (encrypted), RTMPTE (tunneled and encrypted), RTMPS (encrypted over SSL), RTMFPS (layered over UDP instead of TCP)

HTTP Based Streaming protocols

HLS

HLS is an adaptive HTTP-based protocol used for transporting video and audio data from media servers to viewers' screens. Regardless of whether we are watching live video via an app on phone or on-demand content on our TV, chances are that HLS streaming is involved. It supports Apple devices only.

HLS streaming dynamically adapts the resolution to everyone's circumstances. Called adaptive bitrate streaming, this allows broadcasters to deliver high-quality streams to users with outstanding bandwidth and processing power, while also accommodating those lacking in the speed and power departments. Rather than creating one live stream at one bitrate, a transcoder (usually located in the server) is used to create multiple streams at different bitrates and resolutions. The media server then sends the highest-resolution stream possible for each viewer's screen and connection speed.

Creating multiple renditions of a single stream helps prevent buffering or stream interruptions. Plus, as a viewer's signal strength goes from two bars to three, the stream dynamically adjusts to deliver a superior rendition.

- **Audio Codecs:** AAC-LC, HE-AAC+ v1 & v2, MP3
- **Video Codecs:** H.265, H.264
- **Playback Compatibility:** Great (All Google Chrome browsers; Android, Linux, Microsoft, and MacOS devices; several set-top boxes, smart TVs, and other players)
- **Benefits:**
 - Adaptive bitrate and widely supported
 - Unlike the RTMP protocol used in conjunction with Flash player, HLS can easily scale for delivery using ordinary web servers across a global content delivery network (CDN). By sharing the workload across a network of servers, CDNs accommodate viral viewership spikes and larger-than-expected live audiences. They also help improve viewer experience by caching audio and video segments.
- **Drawbacks:** Quality of experience is prioritized over low latency
- **Latency:** 6-30 seconds (lower latency only possible when tuned)
- **Variant Formats:** Low-Latency HLS (see below), PHLS (Protected HTTP Live Streaming)

MPEG-DASH

The Moving Pictures Expert Group (MPEG), an international authority on digital audio and video standards, developed Dynamic Adaptive Streaming over HTTP (DASH) as an industry-standard alternative to HLS. Basically, with DASH you get an open-source option.

- **Audio Codecs:** Codec-agnostic
- **Video Codecs:** Codec-agnostic
- **Playback Compatibility:** Good (All Android devices; most post-2012 Samsung, Philips, Panasonic, and Sony TVs; Chrome, Safari, and Firefox browsers)
- **Benefits:** Vendor independent, international standard for adaptive bitrate
- **Drawbacks:** Not supported by iOS or Apple TV
- **Latency:** 6-30 seconds (lower latency only possible when tuned)
- **Variant Formats:** MPEG-DASH CENC (Common Encryption)

HCL's Take on Streaming Protocols

HLS, MPEG-DASH are currently the most widely used streaming protocols over Http that supports adaptive bitrate media streaming. It is designed and well supports for media broadcasting. It shall be considered when broadcasting live events and sports, where quality is key and requires adapting to different streaming bandwidth. This architecture also requires server component performing encoding that could support for adaptive streaming and distribution component supporting content delivery to the client component. Considering this, HTTP based protocols may not be a suitable option for this presented use case in which we are trying to prototype a peer to peer video streaming.

RTSP, RTMP, Web-RTC are other streaming protocols that are widely been used for long term and available for media streaming. Theoretically RTSP or RTMP are implemented on higher layers and are independent of underlying networking layer and the hardware used. Considering this it would work well on Wi-Gig technologies too. However, practically we may need to further analyze and study its working model on top of Wi-Gig.

Digital Rights Management (DRM) and High-bandwidth Digital Content Protection (HDCP)

Digital rights management (DRM)

DRM tools or technological protection measures (TPM) are a set of access control technologies for restricting the use of proprietary hardware and copyrighted works. DRM technologies try to control the use, modification, and distribution of copyrighted works (such as software and multimedia content), as well as systems within devices that enforce these policies.

Some of the general DRM techniques involve

- Product key verification
- Online activation
- Online verification
- Content scrambling system (CSS)
- Windows Media DRM – Regional lockout
- Tracking through watermarks
- etc.,

High-bandwidth Digital Content Protection (HDCP)

HDCP is a form of digital copy protection developed by Intel Corporation to prevent copying of digital audio & video content as it travels across connections. Types of connections include DisplayPort (DP), Digital Visual Interface (DVI), and High-Definition Multimedia Interface (HDMI), as well as less popular or now deprecated protocols like Gigabit Video Interface (GVIF) and Unified Display Interface (UDI). The system is meant to stop HDCP-encrypted content from being played on unauthorized devices or devices which have been modified to copy HDCP content. Before sending data, a transmitting device checks that the receiver is authorized to receive it. If so, the transmitter encrypts the data to prevent eavesdropping as it flows to the receiver

HDCP uses three systems:

- Authentication prevents non-licensed devices from receiving content.
- Encryption of the data sent over DisplayPort, DVI, HDMI, GVIF, or UDI interfaces prevents eavesdropping of information and man-in-the-middle attacks.

-
- Key revocation prevents devices that have been compromised and cloned from receiving data.

This mechanism governs the transfer of media between devices that originate protected content (Sources) and displaying devices (Sinks). A common use case is a Miracast® user displaying an HDCP-protected movie from their mobile phone or tablet to a television

How DRM Works?

When a user attempts to play back a video, the video player requests a key from a license server. The server determines whether the user and device are authorized, before issuing a license response with a decryption key. The player can then decrypt and play back the content for the user.

For each asset that we want to serve with DRM, we'll need to generate an encryption key, an asset ID, and a key ID. We can generate these keys and IDs yourself, or use the tools provided by the license server to generate them automatically.

We'll ingest these keys and IDs into your license server so that it can be sent to the player, which will use the key to decrypt the content. The keys and IDs, along with a few other parameters, are also used to encrypt and package the content.

Once the content is encrypted and packaged, it needs to be transferred to the origin server or CDN for streaming to your users.

Enabling DRM requires changes to at least three components of your streaming workflow:

1. **Content** - your assets must be transcoded, encrypted, and packaged in formats compatible with the DRM systems you need to support.
2. **Player** - your video player must be able to request a key from a license server and decrypt the video; this may require different players on different platforms.
3. **License Server** - your video player will request decryption keys from a license server every time a piece of content is requested; the license server authenticates and responds to these requests.

OTT Players like Netflix, Prime & DRM Issues

One of the biggest concerns for paid content providers like Netflix is that their content can be illegally leaked, and the number of paid subscribers may decline, or growth may slow.

To prevent such illegal content leakage, Netflix and other OTT services protect the contents by using technologies called “**DRM**” and “**watermarking**”. By default, the original content is encrypted through a process called “**DRM packaging**” and passed to the user. Without the encryption key information used to encrypt the content, the user cannot decrypt it for playback.

With Netflix as an example, the Netflix iOS / Android app allows users to download some TV shows and movie content to their mobile device via the “Save” feature, and to view them offline later. Because the video content is encrypted with DRM, it is impossible for unauthorized users to play it with a general video player even if they copy the content files. For users who have right to use the content, the ‘**encryption key**’ and ‘**usage right information**’ are transmitted separately from the content.

Wired Casting

Casting using a wired interface uses a port from the device and software and drivers already installed for castings.

HDMI and DisplayPort

Both HDMI and DisplayPort can deliver high-definition digital video and high-resolution audio from a source device to a display.

HDMI explicitly supports CEC (Consumer Electronics Control) for controlling entire A/V setups, and an HDMI cable can carry ethernet information (as a variant). DisplayPort supports CEC over an auxiliary channel, but it's rarely if ever implemented, due simply to DisplayPort's faint footprint in the consumer electronics world.

HDMI 2.1 is capable of supporting bit rates up to 48Gbps. DisplayPort 2.0, can handle raw throughput up to 80Gbps (up to 32 Gbps in DisplayPort 1.4). However, as of Nov 2019, DisplayPort 2.0 had not been implemented in any devices; and HDMI 2.1, which makes many of its features optional, has been fully adopted in relatively few real-world products. [1]

For media up-to 4K, HDMI seems to be more suitable over DisplayPort.

HDMI/DisplayPort Features	Common		Latest	
	HDMI 2.0b	DisplayPort 1.4a	HDMI 2.1	DisplayPort 2.0
Number of simultaneous displays	2	4	2	4
Display connector compatibility via adapters:	DVI, VGA	DVI, VGA, HDMI	DVI, VGA	DVI, VGA, HDMI
Single display video resolution (60hz or better/4:4:4)				
Maximum uncompressed	3,840 by 2,160 @60fps 24bpp	5,120 by 2,800 @60fps 24bpp	5,120 by 2,880 @60fps 24bpp	10,240 by 4320 @60fps 24bpp (10K)
Maximum with Display Stream Compression	NA	7,680 by 4,320 @60fps 30bpp	10,240 by 4,320 @60fps 30bpp (10K)	15,360 by 8,460 @60fps 30bpp (16K)
Audio Resolution				
Max sample rate and bit depth	192khz/24-bit	192kHz/24-bit	192kHz/24-bit	192kHz/24-bit
Max audio channels	32	32	32	32
Pass through Audio Resolution				
Max sample rate and bit depth	44.1/16-bit (ARC)	192kHz/24-bit	192kHz/24-bit (eARC)	192kHz/24-bit
Max audio channels (uncompressed)	2	7.1 (times 4)	7.1	7.1 (times 4)
Max audio channels (Compressed)	5.1	Not necessary	Not necessary	Not necessary
Other features				
HDCP (High Definition Content Protection)	2.2 (HDMI 2.0b)	2.2	2.2	2.2
Power delivery?	Rarely used	Thunderbolt 1&2 only	Rarely used	Thunderbolt 1&2 only
Ethernet?	Yes (special cable)	No	Yes (special cable)	No
CEC?	Yes	Possible via Aux channel	Yes	Possible via Aux channel
Free-Sync (aka VESA AdaptiveSync)?	No	Yes	Yes	Yes
HDR?	Static metadata	Static metadata	Dynamic metadata	Dynamic metadata
3D?	Yes	Yes	Yes	Yes

Table 4 Comparison of HDMI and DisplayPort

USB Type-C

With USB 3.1, a corollary standard for a new, updated connector was also introduced: USB Type-C, officially titled—USB-C. Popularly referred to simply as Type-C.

A feature called “Alternate Mode” allows Type-C ports and cables to be used for other transport protocols, including Thunderbolt, DisplayPort, MHL (Mobile High-definition Link, for phones), VirtualLink (for VR headsets), and HDMI. Many devices like smartphones, tablets and Laptops have started adopting the Type-C port. But it’s still available mostly in the high-end devices.

MHL: Mobile High Definition link

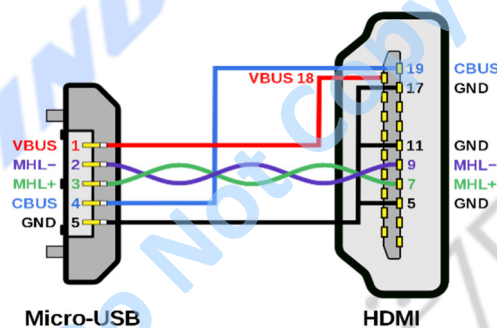


Figure 11 MHL Connectivity

- MHL technology is a HD audio and video interface that connects portable electronics devices such as mobile phones, to HDTVs. The technology allows mobile devices to output full HD resolution. Together with an MHL-to-HDMI bridge, the MHL-enabled mobile device may become a fully compliant HDMI source and can connect to the television’s standard HDMI input port.
- It is an adaptation of HDMI intended for mobile devices such as smartphones and tablets. MHL requires that the HDMI socket be MHL-enabled, otherwise an active adapter (or dongle) is required to convert the signal to HDMI.
- It has the ability to carry uncompressed HDCP encrypted high-definition video, thus being able to carry out lossless transmission.

- There are a total of five pins used in MHL rather than the 19 used in HDMI, namely: a differential pair for data, a bi-directional control channel (CBUS), power charging supply, and ground.
- MHL uses the micro USB port which is already available on many mobile devices. The USB port switches from USB to MHL when it recognizes an MHL-qualified sink (e.g., a TV) detected on the control wire. Also it has a built-in MHL transmitter chip which can recognize an MHL-capable receiver at the other end, and switches into the MHL transmit mode. The MHL transmitter then sends the AV data over two of the five pins on the micro USB port. The control signals are transmitted via one of the other pins.
- The 5 V line is used to charge the mobile device

Table 5 Versions of MHL

MHL 1	• Supports uncompressed HD video up to 720p. • Provides HDCP content protection, and a minimum of 2.5 W (500 mA) power between sink (e.g., TV) and source (e.g., mobile phone) for charging. • Has a built-in Remote Control Protocol (RCP) function allowing a mobile device to manage the playback of its content on the TV.
MHL 2	• Minimum charging supply to 4.5 W (900 mA), with an optional 7.5 W (1.5 A) maximum allowed. • Support for 3D video was also introduced, permitting 720p 60 Hz, and 1080p video modes. • Protocol same as MHL1.
MHL 3	• Supports 4K Ultra HD (3840 × 2160) video, increasing the maximum bandwidth from 3 Gbit/s to 6 Gbit/s. • The maximum charging supply was increased to 10 W (2 A). • Support for compressed lossless audio formats was added with support for Dolby True HD and DTS-HD Master A
Super MHL	• The specification introduces a reversible 32-pin super MHL connector, which (along with USB Type-C) supports a higher charging power of up to 40 W (20 V / 2 A). • Supports 8K Ultra HD (7680 × 4320) 120 Hz High Dynamic Range (HDR) video. • Audio-only mode also available.

Connections:

There are two types of connection, depending on whether the display device directly supports MHL:

1. **Passive Cable**: Passive cables allow MHL devices to connect directly to MHL-enabled TVs (i.e. display devices or AV receivers with an MHL-enabled HDMI port) while providing charging power upstream to the mobile device. Other than the physical connectors, no USB or HDMI technology is being used. Exclusively MHL signaling is used through the connectors and over the cable.
2. **Active adapter**: With an active adapter, MHL devices are able to connect to HDMI display devices that do not have MHL capability by actively converting the signal to HDMI. These adapters often feature an additional Micro-USB port on them to provide charging power to the mobile device because standard HDMI ports do not supply sufficient current.

Solutions Available in The Market

MOKIN – Extend your Screen Wirelessly

MOKIN lets you extend your screen wirelessly. Designed in two parts, a transmitter, and a receiver, MOKIN lets you connect a device to a display without running a clunky, ugly, entangled cable between the two like an HDMI cable. The transmitter and receiver let you extend your screen wirelessly and seamlessly, offering high-definition images with zero latency. MOKIN's transmitter runs on a USB-C port.



Figure 12 MOKiN - Extend your screen wirelessly

This means you can easily connect your laptop, tablet, or smartphone to an external display or a projector without worrying about dongles and such. The receiver end has standard display ports on it, making it compatible with most TV screens or projectors.

The transmitter unit has its own USB hub, letting you plug pen-drives or SD cards into it and access it right through your device. The transmitter and receiver communicate using low-latency Wi-Fi, and can easily work over distances as long as 10 meters, beaming 1080p content without any problems. The receiver unit has both VGA and HDMI ports, as well as a 3.5mm headphone jack, so not only can you plug a variety of device in, you can use two simultaneous displays, as well as an external speaker.

Current Hardware Trends

Casting Tech

Chromecast Technology

Chromecast is a line of digital media players developed by Google. The devices, designed as small dongles, enable users with a mobile device or personal computer to play Internet-streamed audio-visual content on a high-definition television or home audio system through mobile and web apps that support the Google Cast technology. Alternatively, content can be mirrored from the Google Chrome web browser running on a personal computer, as well as from the screen of some Android devices.

Chromecast is managed through the Google Home app, which enables users to set up new devices and configure existing ones (such as specifying which "Backdrop" images are shown when no other content is cast). Users can also search for streaming content that is available on installed Google Cast-enabled apps. The app manages other Google Cast-supported devices, including the Google Home smart speaker.

Chromecast uses the mDNS (multicast Domain Name System) protocol to search for available devices on a Wi-Fi network. Chromecast previously used the DIAL (Discovery and Launch) protocol, co-developed by Netflix and YouTube.

Model Comparison

Model	Chromecast(1 st generation)	Chromecast(2 nd generation)	ChromecastUltra	Chromecast(3 rd generation)
Storage	2 GB	256 MB	-	-
Display	1080p @ 30fps or 720p @ 60fps	1080p @ 30fps or 720p @ 60fps	4K Ultra HD High dynamic range (HDR10, Dolby Vision)	1080p @ 60fps

Connectivity	HDMI (can use CEC) Wi-Fi (802.11 b/g/n @ 2.4 GHz) Ethernet (with optional USB power adapter)	HDMI (can use CEC) Wi-Fi (802.11 b/g/n/ac @ 2.4/5 GHz) Ethernet (with optional USB power adapter)	HDMI (can use CEC) Wi-Fi (802.11 b/g/n/ac @ 2.4/5 GHz) Ethernet (with included USB power adapter) ^[31]	HDMI (can use CEC) Wi-Fi (802.11 b/g/n/ac @ 2.4/5 GHz) Ethernet (with optional USB power adapter)
Power	Micro-USB (power adapter or USB port)	Micro-USB (power adapter or USB port)	Micro-USB (power adapter required)	Micro-USB (power adapter or USB port)
Memory	512 MB RAM	512 MB RAM	-	-

Table 6 Comparison of Chromecast Models

Miracast Technology

Miracast is a standard for wireless connections from devices (such as laptops, tablets, or smartphones) to displays (such as TVs, monitors, or projectors), introduced by the Wi-Fi Alliance. It can roughly be described as "HDMI over Wi-Fi", replacing the cable from the device to the display.

Miracast employs the peer-to-peer Wi-Fi Direct standard. It allows sending up to 1080p HD video (H.264 codec) and 5.1 surround sound (AAC and AC3 are optional codecs, mandated codec is linear pulse-code modulation – 16 bits 48 kHz 2 channels).[4] The connection is created via WPS and therefore is secured with WPA2. IPv4 is used on the Internet layer. On the transport layer, TCP or UDP are used. On the application layer, the stream is initiated and controlled via RTSP, RTP for the data transfer.

Both devices (the sender and the receiver) need to be Miracast certified for the technology to work. However, to stream music and movies to a non-certified device, Miracast adapters are available that plug into HDMI or USB ports.

Functionality:

The technology was promoted to work across devices, regardless of brand. Miracast devices negotiate settings for each connection, which simplifies the process for the users. In particular, it obviates having to worry about format or codec details. Miracast is "effectively a wireless HDMI cable, copying everything from one screen to another

using the H.264 codec and its own digital rights management (DRM) layer emulating the HDMI system". The Wi-Fi Alliance suggested that Miracast could also be used by a set-top box wanting to stream content to a TV or tablet. Screen mirroring with wired connection and wireless connection has been huge easier. There are different types of apps available in app store and play store which will mirror our screen with wired or wireless connection. HDMI to HDMI screen mirroring is the easiest way.

Display resolution	Types of media streamed	OS support	Protocols
1280 x 720p30	Miracast can stream videos that are in 1080p, media with DRM such as DVDs.	Windows, Android	TCP or UDP, RTSP, RTP

Table 7 Features of Miracast

Phones

Specifications	Asus Zenfone 4 pro/ LeEco Le Max	Asus ROG phone 2018 Asus ROG phone2 2019	Samsung Galaxy S10 2019
			
Wi-Fi Standards	802.11 a/b/g/n/ac/ad	802.11a/b/g/n/ac/ad	802.11 a/b/g/n/ac/ax
Chipsets	Qualcomm Snapdragon 810/835	Qualcomm Snapdragon 845	Qualcomm Snapdragon 855
Remarks	Poor performance in terms of Power consumption, hardware resource usage and antenna placement	Best ad compatible smartphone in the market. Has multiple antennas, large battery capacity and an efficient cooling mechanism primarily targeted towards heavy gaming	Wi-Fi 6 capable

Table 8 Phones with 802.11ad/ay support

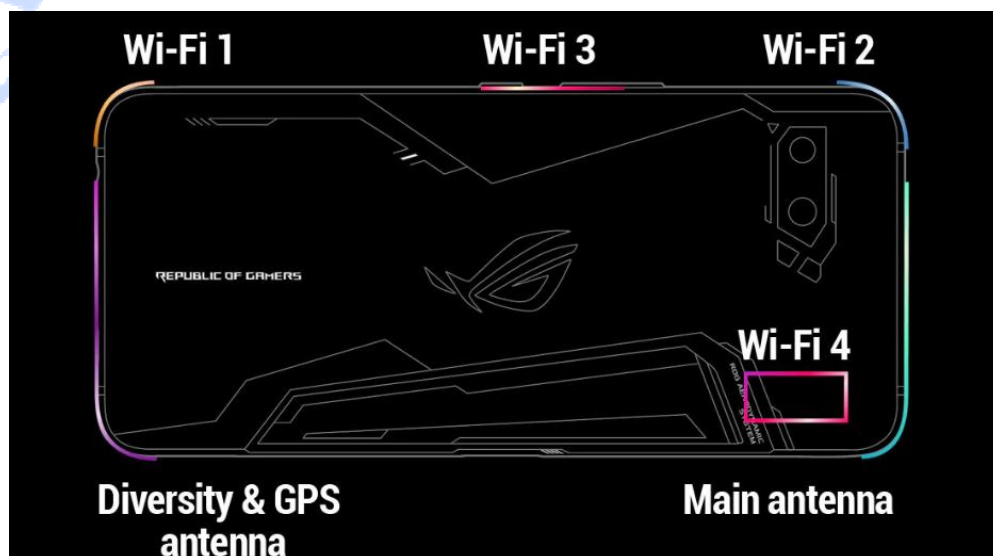


Figure 13 Multi-Antenna in ASUS ROG Phones

-
- By 2019, the flagship phones of Samsung and Apple have been equipped with Wi-Fi 6
 - In May 2020, Qualcomm announced the launch of the first batch of Wi-Fi 6E chips, suitable for routers and mobile phones.
 - Wi-Fi 6E chips supporting mobile phones is expected to be concentrated on the upcoming Qualcomm Snapdragon series

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ANRC

EK1HMC6350 60Ghz Evaluation Kit and Software (Analog Devices)

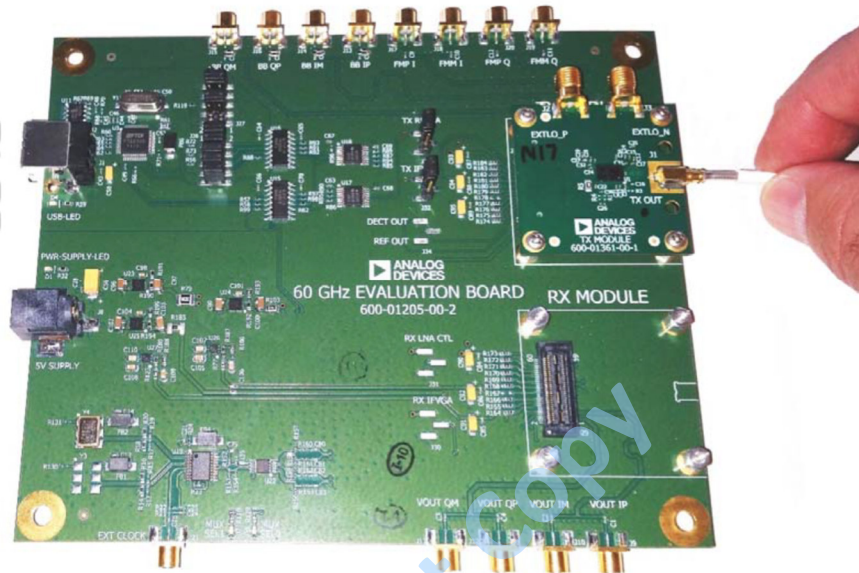


Figure 14 HMC6300 Assembly with the Motherboard Populated with the TX Module

FEATURES

1. 60 GHz evaluation kit designed for quick evaluation of the HMC6300 and HMC6301 transmit/receive chipset
2. Demonstration platform allows easy integration of the HMC6300 and HMC6301 radio link with a baseband modem for over the air transmission

APPLICATIONS

1. Point to point millimeter wave radio development
2. Wireless Gigabyte Alliance/IEEE 802.11ad radio
3. 60 GHz RF sniffing
4. 60 GHz signal generator
5. Modem development
6. Radar/high resolution Imaging

EVALUATION KIT CONTENTS

1. 1 HMC6300 Tx 60 GHz transmitter motherboard/module assembly with a MMPX interface
2. 1 HMC6301 Rx 60 GHz receiver motherboard/module assembly with a MMPX interface
3. 2 hand formable 2 inch coaxial cables with a male MMPX connector on one end and a 1.85 mm male connector on the other end
4. 1 HMC6300 motherboard/TX module with crystal, USB interface, supply regulators, MCX connector and differential I/Q interface
5. 1 HMC6301 motherboard/RX module with crystal, USB interface, supply regulators, MCX connector and differential I/Q interface
6. 2 wall-mount power supplies and 2 USB 2.0 male/male 6 foot cables
7. 8 phase matched MCX to SMA cables for baseband interface on both motherboards
8. Evaluation software CD-ROM containing the HMC6300/HMC6301 Graphical User Interface software GUI

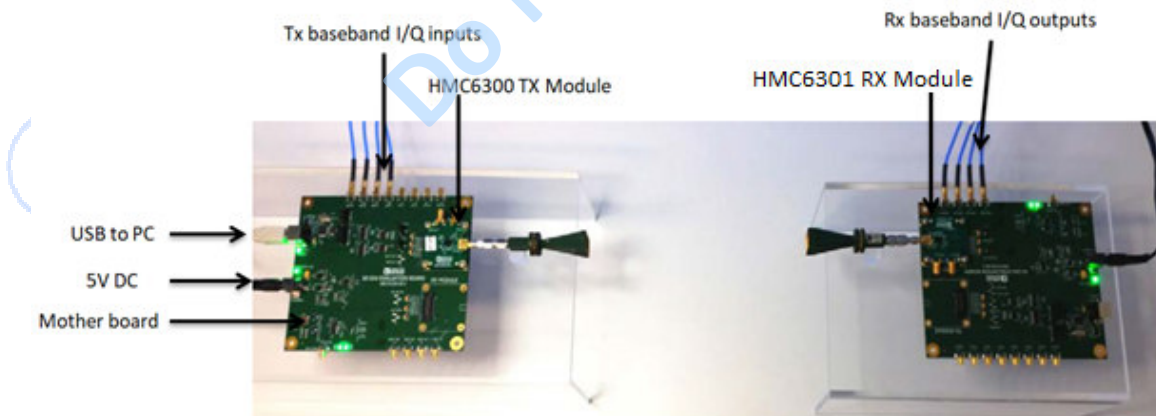


Figure 15 Typical HMC6300 and HMC6301 60GHz Radio Link with USB, DC Power, and I/Q Cables

Tx Setup	Rx Setup
The baseband source can be test equipment or a modem that generates a baseband tone or modulated I/Q signal. Connect the baseband inputs on the motherboard to the baseband I/Q source as follows: Motherboard BB_IP connects to the I positive baseband output Motherboard BB_IM connects to I negative baseband output Motherboard BB_QP connects to Q positive baseband output Motherboard BB_QM connects to Q negative baseband output	Attach the other end of the baseband cables to baseband test equipment or a modem that processes a baseband tone or modulated I/Q signal. Connect the baseband inputs on the motherboard to the baseband I/Q source as follows: Motherboard VOUT_IP connects to I positive baseband input Motherboard VOUT_IM connects to I negative baseband input Motherboard VOUT_QP connects to Q positive baseband input Motherboard VOUT_QM connects to Q negative baseband input

Table 9 Tx & Rx Setup in K1HMC6350 60Ghz Evaluation Kit

SOFTWARE/GUI CONFIGURATION:

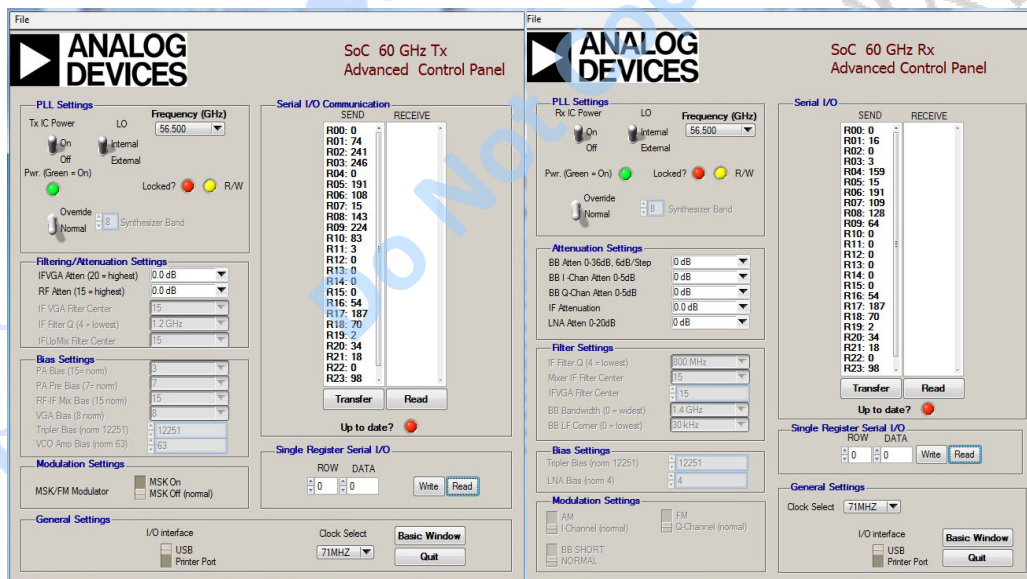


Figure 16 Screenshot of the K1HMC6350 60Ghz Evaluation Kit Software

1. The EK1HMC6350 evaluation kit allows users to set up a half-duplex, 60 GHz millimeter wave link using standard baseband analog interfaces
2. There are two motherboards in the kit: an HMC6300 motherboard with the TX module location populated and the HMC6301 motherboard with the RX module location populated.
3. We can set parameters like carrier frequency, the IF and RF gains, modulations etc. using the GUI as shown above.

Display Hardware

Desirable Features:

- UHD 4K (3840 x 2160) Resolution
- Wired/Wireless Connectivity.
 - HDMI/USB
 - 802.11 ac/ay/ad
- Multi point capacitive touch screen
- Lightweight, slim dimension



Figure 17 An example of an economy-class displays from Qatar Airways

We can develop a custom-made display system using commercially available 4k portable screens. The screen will be controlled by a Linux system which will run the software to communicate with the passenger's device.

A commercially available 4K touchscreen portable display, “CF015xT - 4K Ultra-slim Touch Portable Display” from C-Force compared to 2 existing Inflight Entertainment Displays:

Specification	CF015xT - 4K Ultra-slim Touch Portable Display, C-Force	Passenger Entertainment + Information System, Moving Terrain	15.3" High Definition Seat Display Unit, AeroVision Avionics Inc.
Screen size	15.6 inch	10.4 inch	15.3 inch
Resolution	3840*2160 60Hz	1024*768	1280*768
Viewing angle	178°	170°	170°
Average brightness	300cd/m ²	130cd/m ²	300cd/m ²
Touch function	Yes, capacitive	No	No
Size	373mm*219mm	234mm*181mm	388.6mm*249mm
Thickness	4.8mm~9.5mm	12mm	42.4mm
Input voltage	USB-PD 5V~20V	24/28V, 18W	28V, 31W
IO type	mini-HDMI / USB-C / 3.5mm headphone jack / micro USB 2.0	VGA/DVI/Composite video	HD SDI
Built-in speaker	Yes	-	-
Weight	741g	870g	3628g

Table 10 Comparison of a portable 4K display with existing Inflight Entertainment Displays

Associated Cost

Component	Approximate Cost (\$)
CF015xT - 4K display	270
802.11ad receiver hardware	350
Linux System	50
Internal cables	10
Total	680

Table 11 Approximate cost of display components

The cost of the Infotainment System will be below 700 dollars if we are building 1 unit from commercially available components.

“ "Next to the engines, it's the second most expensive item on an aircraft," said Lori Krans, spokeswoman for Thales, one of the world's largest IFE makers. Industry analysts say that the cost to retrofit an aircraft run from US \$2 million to US \$5 million.” [3]

IEEE 802.11ad (Wi-Gig)

The IEEE 802.11ad amendment addresses the challenge of efficient use of the vast 60 GHz spectrum and a transition from omnidirectional to directional wireless medium usage, bringing multi-gigabit-per-second throughput and new application scenarios to Wi-Fi users. These new uses include instant wireless synchronization, high-speed media file exchange between mobile devices without fixed network infrastructure, and wireless cable replacement (e.g., wireless casting).

60 GHz propagation behavior shows increased signal attenuation. At a typical IEEE 802.11ad range of 10 m, additional attenuation of 22 dB compared to the 5 GHz band is predicted by the Friis transmission equation, resulting from the frequency-dependent difference in antenna aperture. In contrast, oxygen absorption plays a minor role over short-range distances, even though it peaks at 60 GHz.

Furthermore, 60 GHz communication is characterized by a quasi-optical propagation behavior where the received signal is dominated by the line of sight (LOS) path and first order reflections from strong reflecting materials. As an example, metallic surfaces were found to be strong reflectors and allow non-LOS (NLOS) communication. Concrete materials, on the other hand, cause additional large signal attenuation and can easily create a blockage.

Thus, 60 GHz communication is more suitable to in-room environments where sufficient reflectors are present.

Directional Communication

The IEEE 802.11ad standard defines a directional communication scheme that takes advantage of beamforming antenna gain to cope with increased attenuation in the 60 GHz band. With quasi-optical propagation behavior, low reflectivity, and high attenuation, beamforming results in a highly directional signal focus. Based on this behavior, the standard introduces a novel concept of “virtual” antenna sectors that discretize the antenna azimuth. IEEE 802.11ad sectors can be implemented either using precomputed antenna weight vectors for a phased antenna array or equipping a system with multiple directional antenna elements. In both cases, the wavelength in the

millimeter range allows antenna form factors significantly smaller than those of legacy Wi-Fi at 2.4/5 GHz.

A sector focuses antenna gain in a certain direction. Communicating nodes have to agree on the optimal pair of receive and transmit sectors to optimize signal quality and throughput. This process, referred to as beamforming training, takes advantage of the discretized antenna azimuth that reduces the search space of possible antenna array configurations. After a first sector matching, a second beam training stage allows further refinement of the found sectors. During this stage, antenna weight vectors that vary from predefined sector patterns can be evaluated to further optimize transmissions on phased antenna arrays. While in general higher antenna gain is desirable, it imposes stronger directionality and a higher number of narrow antenna sectors. This increases coordination overhead to adapt the antenna steering between communicating nodes, and it has been shown that link budget loss by misalignment increases with directionality.

The figure shows an example for two nodes communicating over virtual IEEE 802.11ad sectors. The highlighted selection of sectors that matches the LOS direction may offer the optimum link quality in the absence of blocking obstacles.

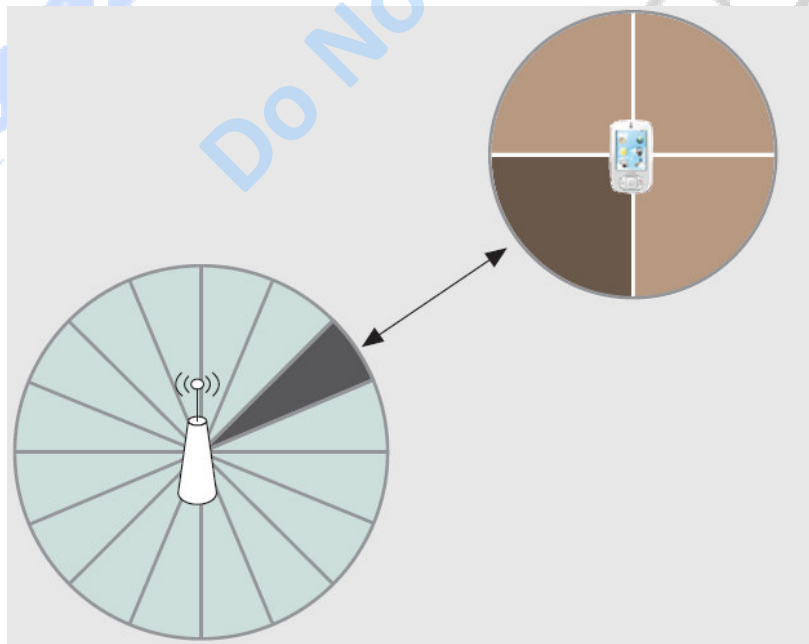


Figure 18 Virtual Antenna Sectors

Directional Multi-Gigabit (DMG)

The fields of a DMG packet are as follows:

- STF - The short training field, which is used for synchronization.
- CE - Channel estimation field, which is used for channel estimation.
- Header - The signalling field, which the receiver decodes to determine transmission parameters.
- Data - The data field, which carries the user data payload.
- AGC Subfields - Optional automatic gain control (AGC) subfields, used for beam refinement.
- Training Subfields - Optional training subfields, used for beam refinement.

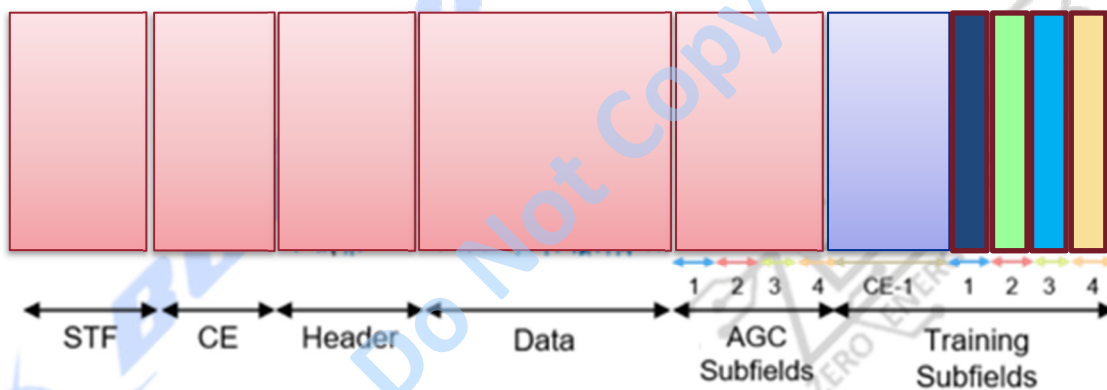


Figure 19 The DMG Packet Format

IEEE 802.11ad defines the directional multi-gigabit (DMG) transmission format operating at 60 GHz. To overcome the large path loss at 60 GHz, 802.11ad standard is designed to support directional beamforming. By using phased antenna arrays you can apply an antenna weight vector (AWV) to focus the antenna pattern in the desired direction by applying a phase shift to each element to steer the transmission. The quality of a communication link can be improved by appending optional training fields to DMG packets, and testing different AWVs at the transmitter or receiver. This process is called beam refinement.

802.11ad Waveform Generation with Beamforming - MATLAB

This example simulates the transmission of one Directional Multi-Gigabit packet from a transmitter to a receiver placed at an angle with respect to a set direction. The example has been used for studying the magnitude of the wave at different angles for various typical device configurations found in literature.

Beamforming Specification:

- The transmitter antenna pattern is configured as a 16-element uniform linear array with half-wavelength spacing.
- *WLAN Toolbox* is used to generate a waveform
- *Phased Array System Toolbox* is used to apply the Antenna Weight Vectors and evaluate the magnitude of the resultant plane wave in the direction of a receiver.

Some key parameters that can be changed are:

- The angle at which the receiver is placed wrt to (0°), receiverAz
- The angles for the training fields, (trnAngle)
- Number of training fields (dmg.TrainingLength)
- Number of Array elements in the transmitter antenna (N)
- The preamble angle
- Various angles for the training fields

The plotDMGWaveform function is used to plot the magnitude of beamformed plane wave.

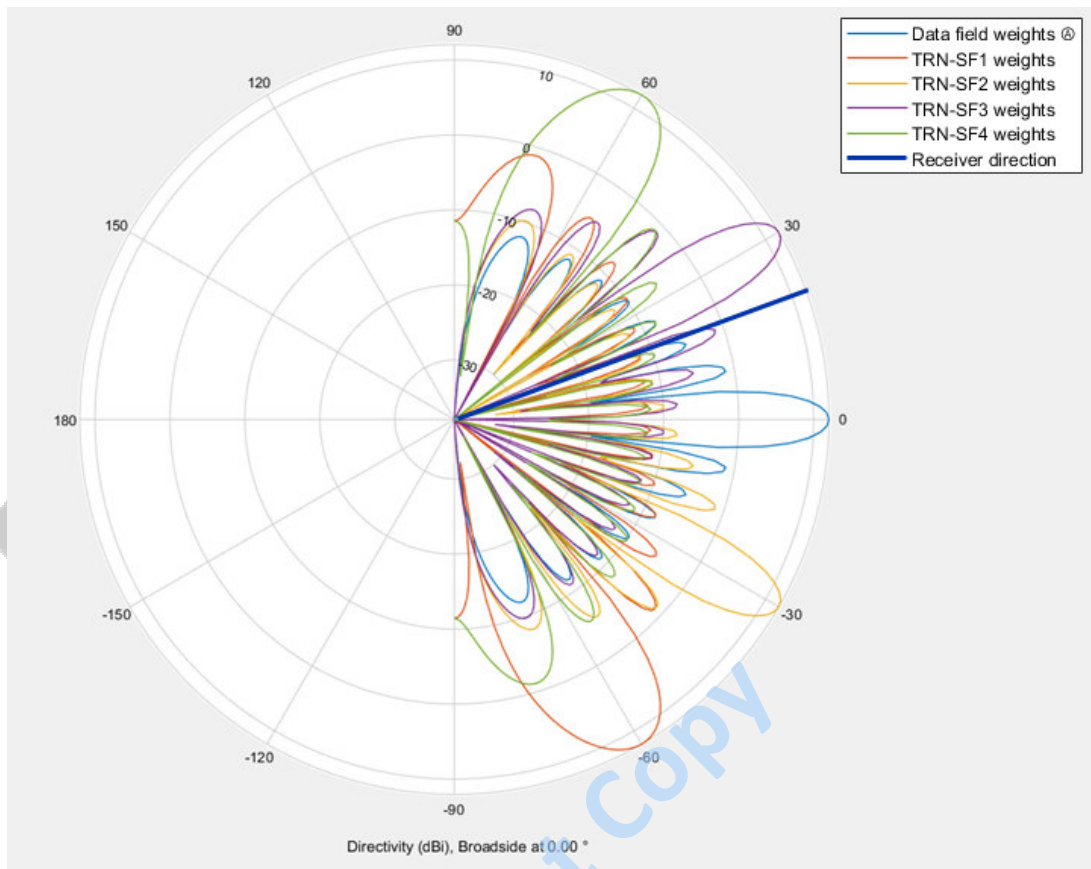


Figure 20 Directivity of Various Fields

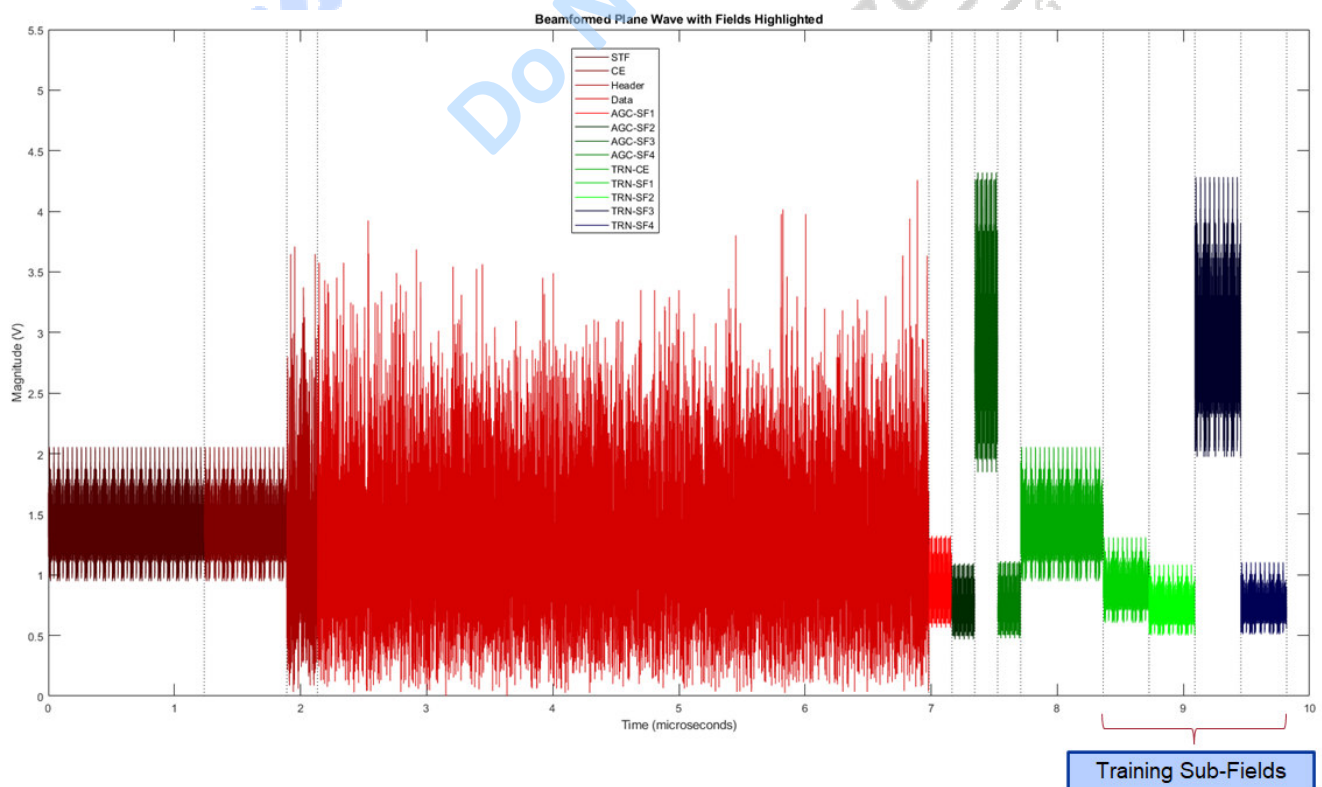


Figure 21 Magnitude of the beamformed plane wave

Observations

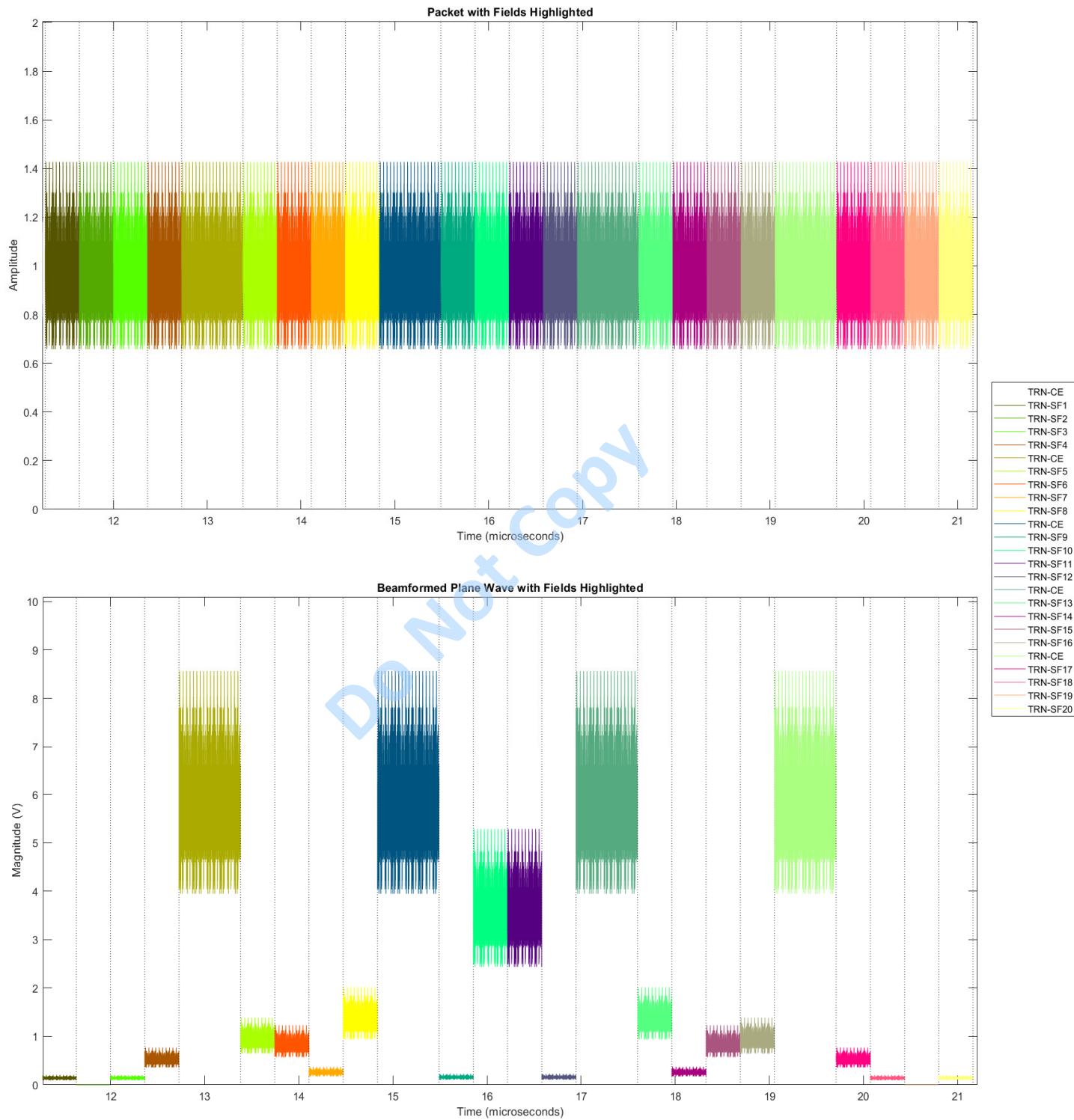


Figure 22 A basic DMG packet compared with the transmitted beamformed plane wave

The above figure shows the plot of the training subfields of a basic DMG packet and the transmitted beamformed plane wave. Note that the scales of the two plots are not to

be compared. According DMG packet format, there is one TRN-CE field for every 4 training subfields.

It is clear from the simulations that the magnitude is the highest when the beam is directed directly towards the receiver. The magnitudes drastically fall as the angle is moved away from the receiver. This seems to be an advantage as we will see minimum interference between two passengers when they are casting over 802.11ad Wi-Fi to their respective seat back displays.

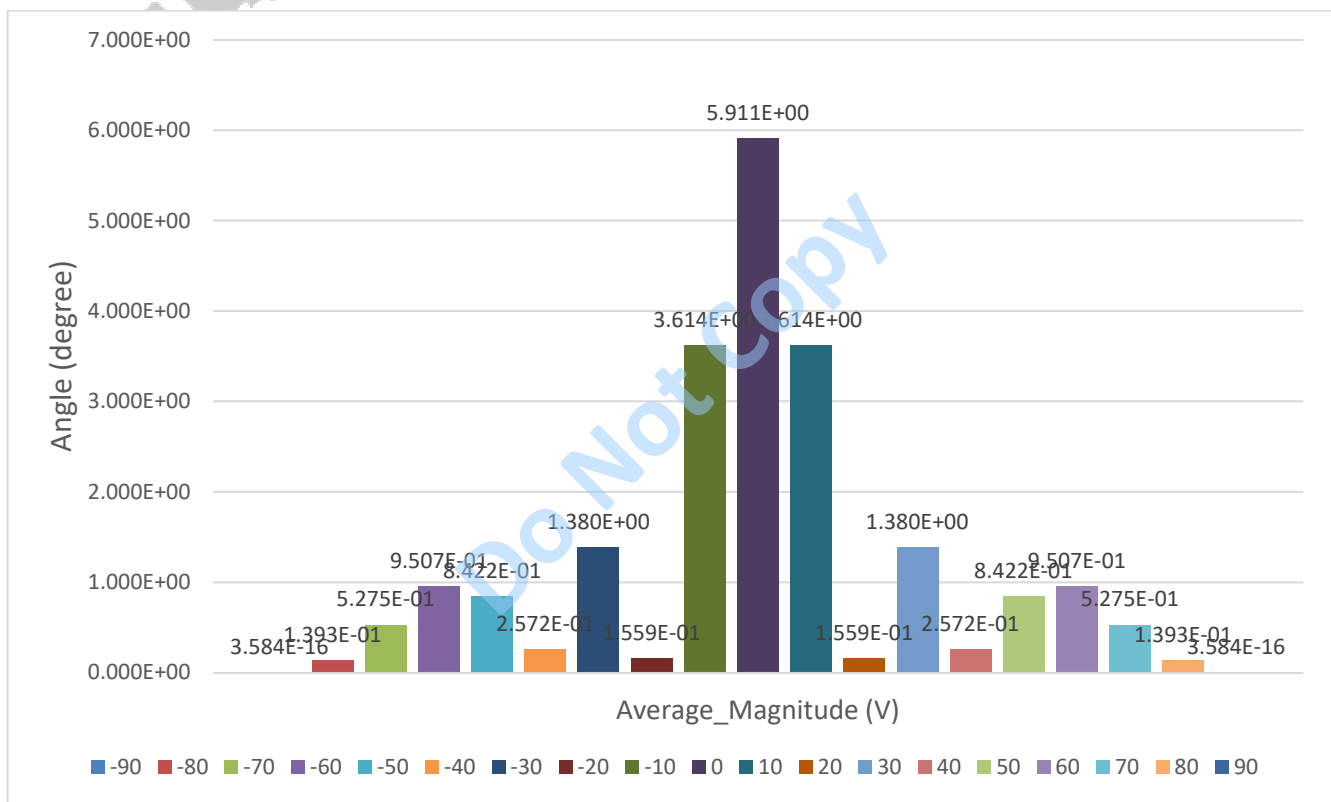


Figure 23 Average magnitude of beamformed plane wave at various angles

The above figure shows the average magnitude of beamformed plane wave at various angles. The transmitter has 6 antenna elements resembling a small device like a mobile phone. This assumption is based on the fact that the Millitronic WiGiG USB Adapter has 6 elements on 1 side and 4 on the other side of its antenna.

The angles are in the step size of 10° across 180° with the boresight being 0°. I.e., this is achieved using beamforming while the antenna remains fixed. The receiver is along 0°.

Further investigation is being done to determine the data rate, effect of directivity etc.

IEEE 802.11ax Wi-Fi 6

The amount of available unlicensed spectrum has remained static in the past 20 years. The United States Federal Communications Commission (FCC) has announced the availability of the 6 GHz frequency band for unlicensed spectrum use. This creates new opportunities for tremendous innovation and the continued success and economic growth enabled by Wi-Fi.

Motivation for 802.11ax Wi-Fi 6E

1. Exploit the channels available in the 6GHz bands
2. Perform end-to-end simulation of multiple transmitter receiver stations in cabin environment to obtain PER and throughputs
3. Plan the hardware specification for deployment in seat back displays

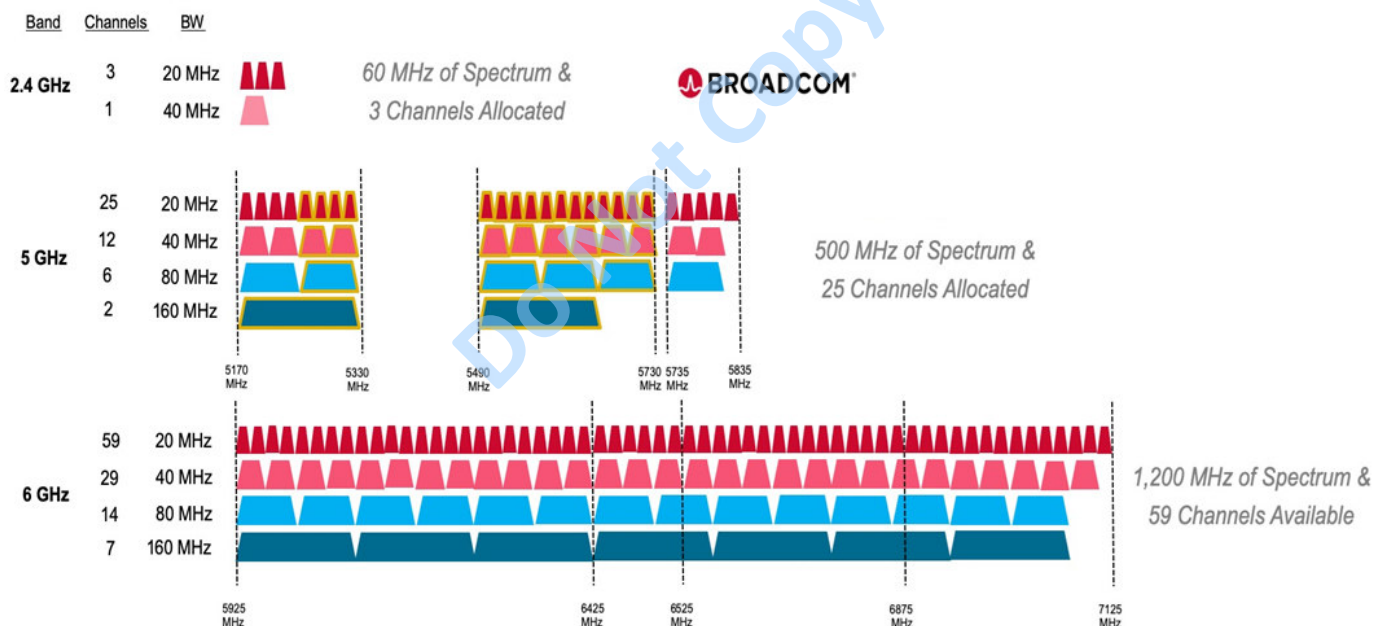


Figure 24 Wi-Fi Bands at 2.4, 5 & 6 GHz

Maximum theoretical speed of Wi-Fi 6E:

- mobile phone chip 3.6Gbps
- router chip 5.4Gbps to 10.8Gbps

Wi-Fi 6E brings Wi-Fi® into 6 GHz

Features	Benefits
More, contiguous spectrum	Gigabit speeds
Wider channels	Extremely low latency
Less interference	High capacity

802.11ax Simulation in MATLAB

Here we perform a PHY-focused system-level simulation for IEEE® 802.11ax. The simulation comprises of:

- Simulation scenario
- Radio characteristics
- Large-scale fading model using TGax
- Estimating the packet error rate of the 802.11ax network

We split the simulations in two parts.

Part (A) validates the simulation scenario, radio characteristics, and large-scale fading model by comparing against published calibration results.

Part (B) estimates the packet error rate of the 802.11ax network by simulating individual links between active nodes under a basic clear channel assessment scheme.

Simulation scenario:

The simulation scenario consists of Boeing 777x economy class seating arrangement. Refer to figures below. The space available between each screen is represented by a cell. Each cell has an access point (AP) on the seat back display and one station (STA) for the user's device. A process referred to as 'dropping' nodes is used to create a basic service set (BSS) per cell which is randomly assigned one of three channels. The simulation scenario specifies a large-scale path loss model based on the distance between nodes. The TGax evaluation methodology for 'PHY System Simulation' is followed for this example:

1. APs and STAs are randomly 'dropped' within the scenario.
2. For each pair of nodes, the large-scale path loss is calculated.
3. One or more 'transmission events' is performed.

Before simulation, the transmitter and receiver sites are generated and assigned cell names for ease of reference. A transmitter (AP) and receiver (STA) pair is assigned per cell with both being isotropic.

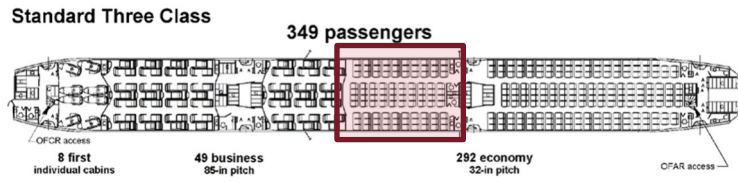


Figure 25 Boeing 777-9 Economy Class

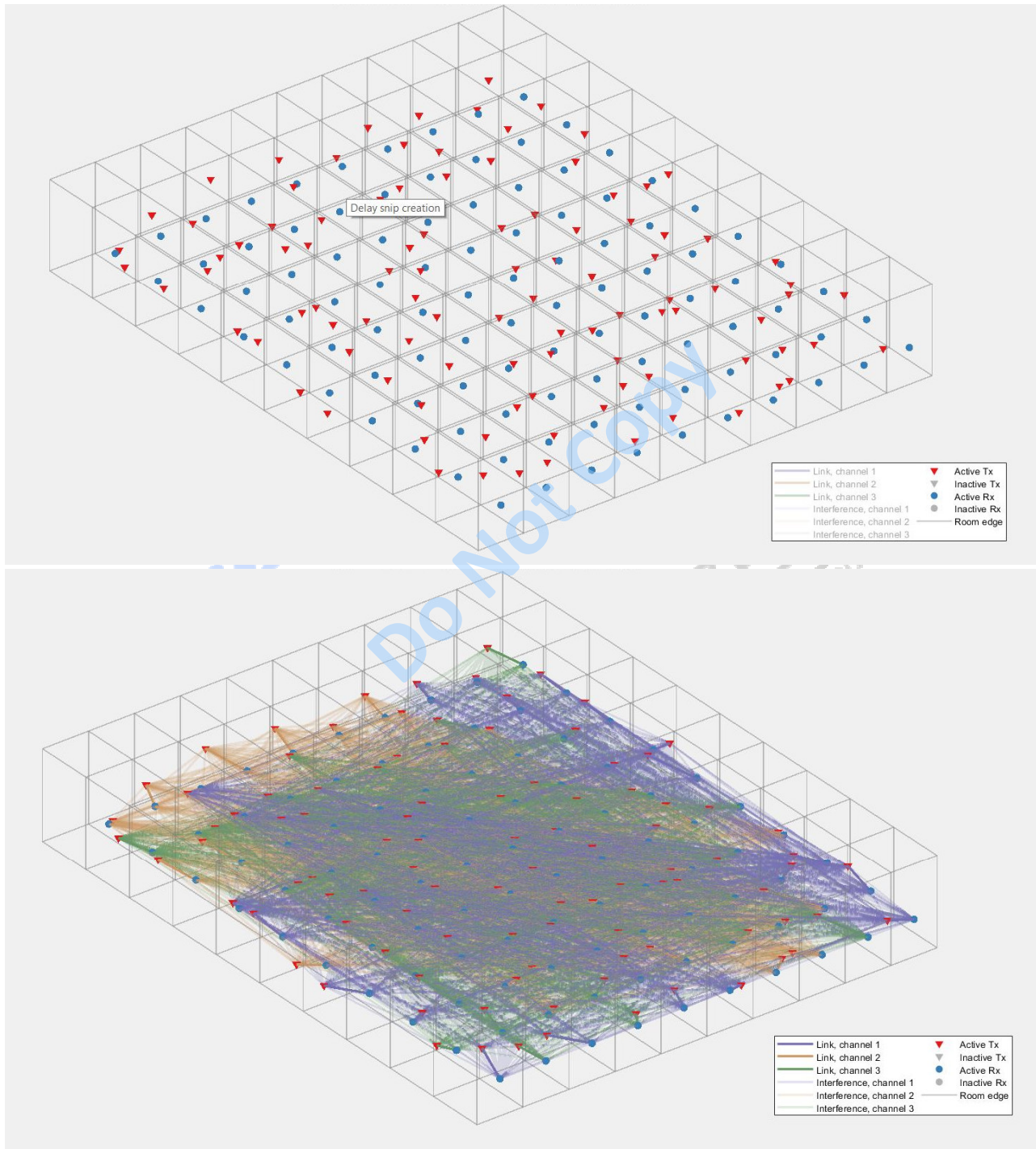
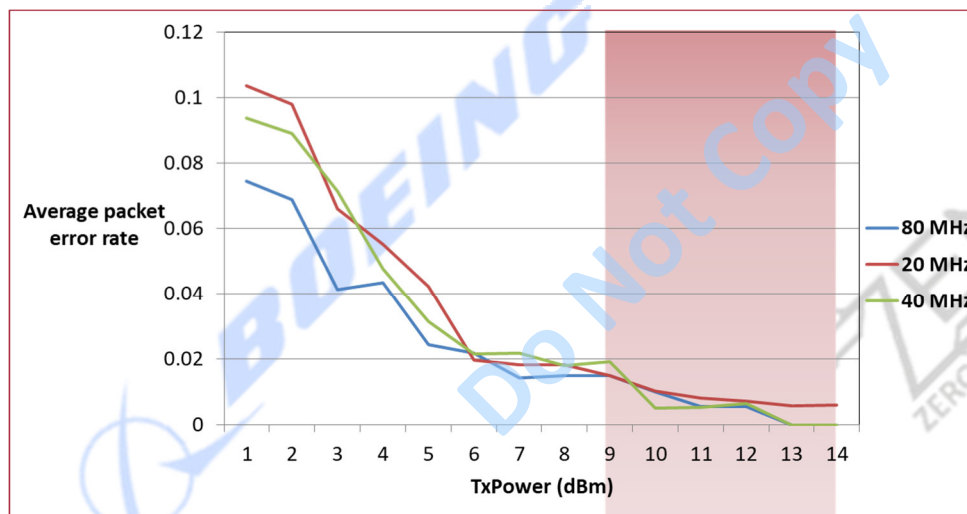


Figure 26 802.11ax Links established over 3 channels between 100 pairs of Tx-Rx

Simulation parameters:

Parameter	Details
Tx Power	Transmission power for mobile wifi transmitters falls between 9 to 14 dBm
Channel Bandwidth	We repeat the simulations for 20,40,80 and 160Mhz channel bandwidths
Transmitter frequency	5 and 6Ghz carriers
Number of cells in [x,y] directions	A *insert length * long section of cabin is taken which covers * insert cells*
Dimensions of each cell (m)	0.6 x 0.8128 x 1.5

Observations

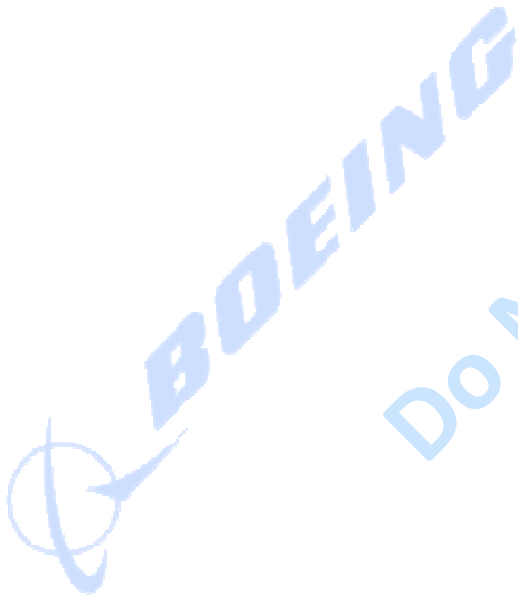


- We see that for transmission powers between 9 to 14 dBm, the average packet error rates converge.
- Wi-Fi 6 has a large number of 20 GHz channels, which allows us to accommodate a large number of devices in the same physical space where as a limited number of higher-throughput 80Ghz channels could be managed by a simpler channel allocation algorithm.

WLAN Controllers

a

WEG



Do Not Copy



Proposed Work

1. Test Setup

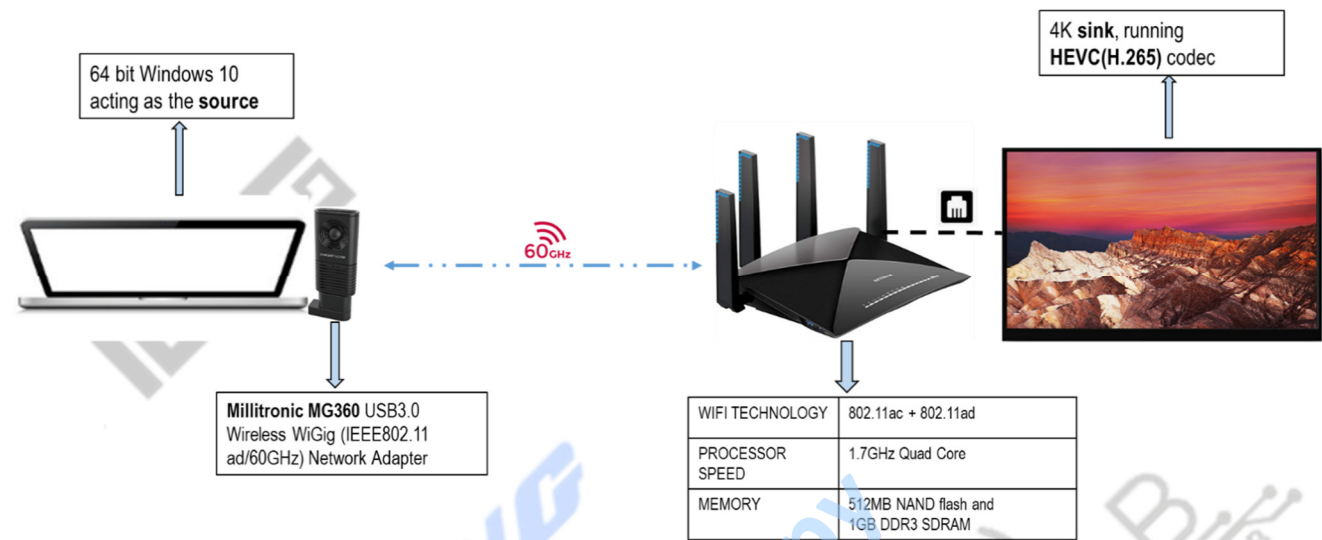


Figure 27 Test setup

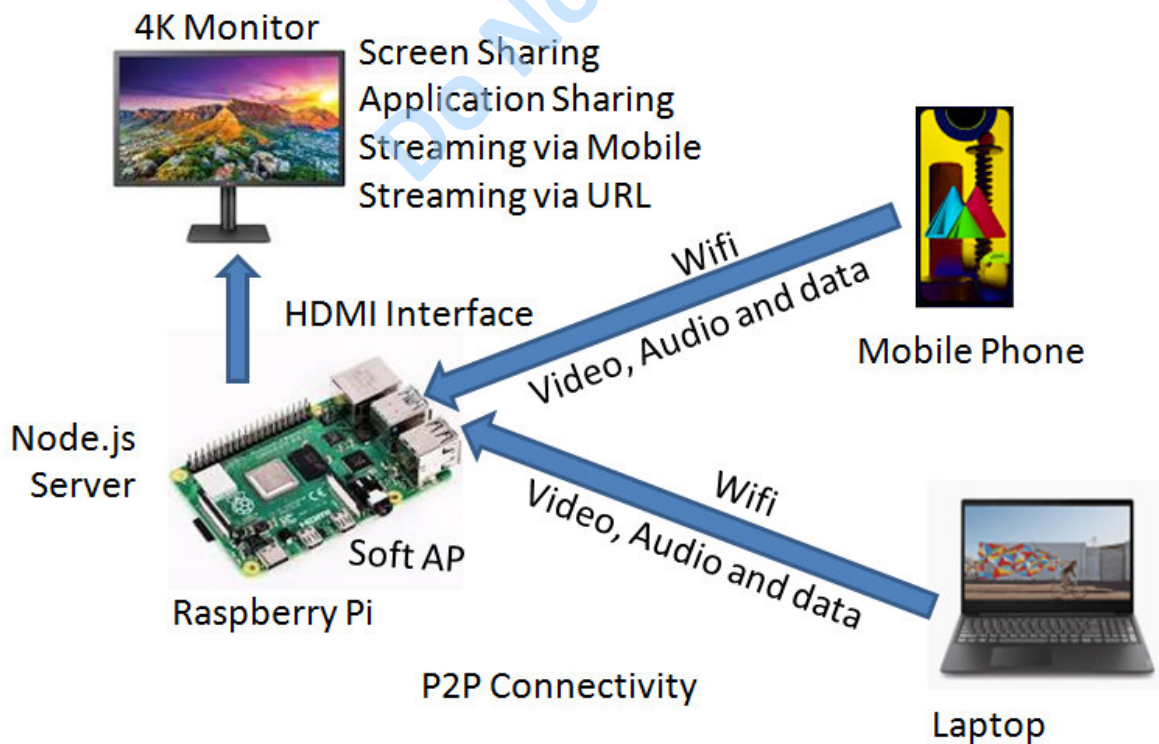


Figure 28 Work Setup - Wipro

1. The transmitter can take input from any MHL enabled device using USB type C or micro USB.
2. User devices are used to directly control media like in a standard MHL link.

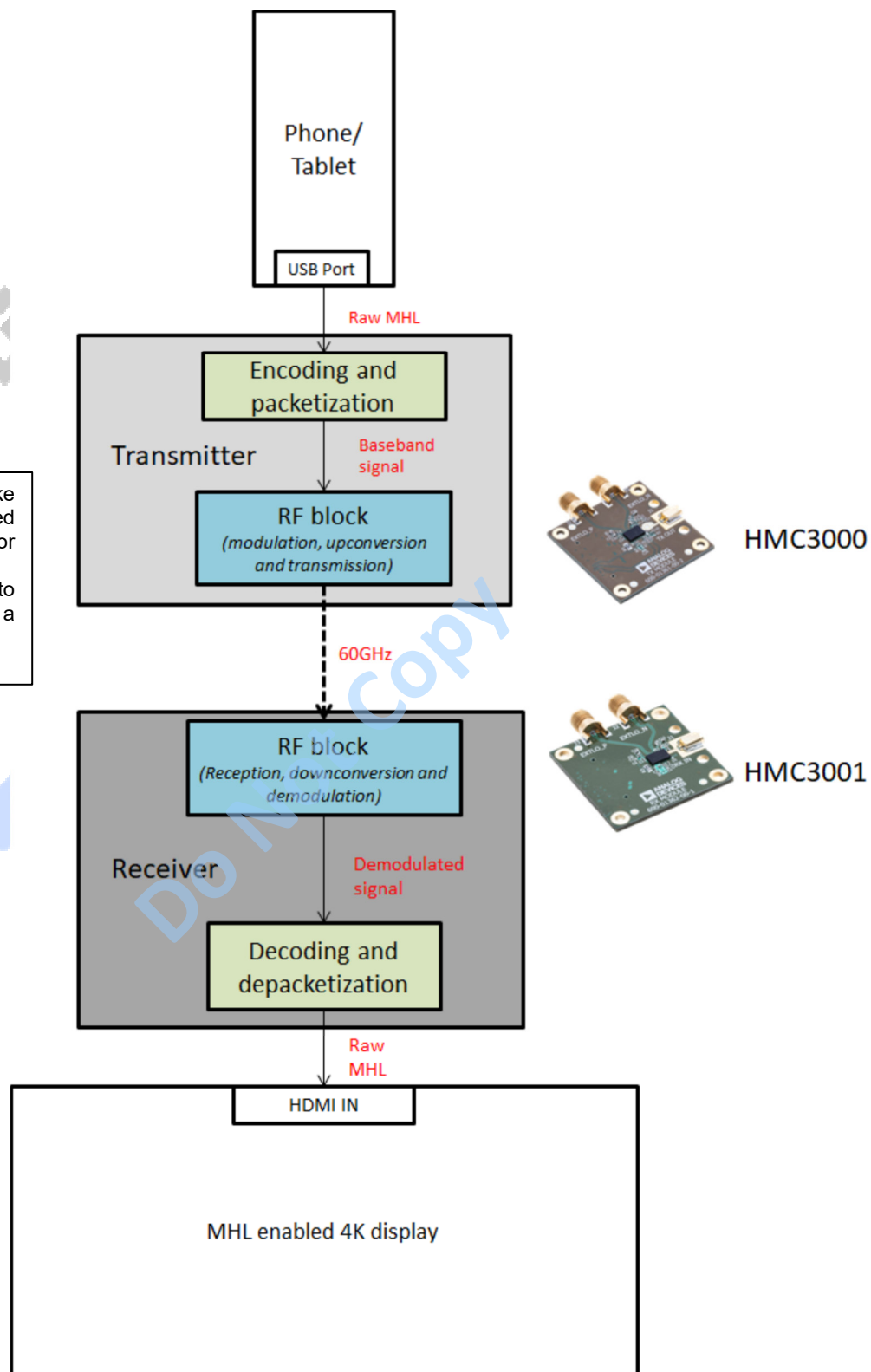
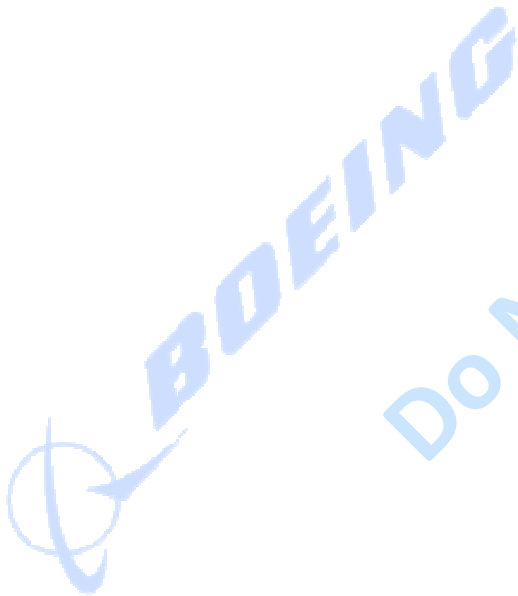


Figure 29 Wireless MHL using 60GHz Analog Device modules

-
2. Designing the Display unit
 3. Wired casting solution (using test setup)
 4. Wireless casting solution (using test setup)
 - a) Wireless peer-to-peer
 - b) Wireless access point
 5. Application Software
- DLNA/VLC/...



Appendix

MIMO

Multiple-Input Multiple-Output (MIMO) is a wireless technology that uses multiple transmitters and receivers to transfer more data at the same time. All wireless products with 802.11n support MIMO. The technology helps allow 802.11n to reach higher speeds than products without 802.11n.

To implement MIMO, the station (mobile device) or the access point (AP) must support MIMO. For optimal performance and range, both the station and the AP must support MIMO.

MIMO technology uses a natural radio-wave phenomenon called multipath. With multipath, transmitted information bounces off walls, ceilings, and other objects, reaching the receiving antenna multiple times at different angles and slightly different times. In the past, multipath caused interference and slowed down wireless signals. With multipath, MIMO technology uses multiple, smart transmitters and receivers with an added spatial dimension, increasing performance and range.

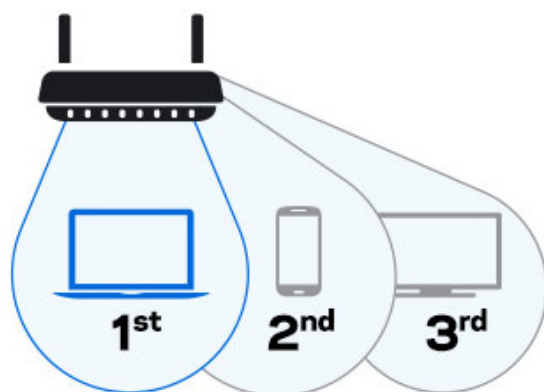
MIMO increases receiver signal-capturing power by enabling antennas to combine data streams arriving from different paths and at different times. Smart antennas use spatial diversity technology, which puts surplus antennas to good use. When antennas outnumber spatial streams, the antennas can add receiver diversity and increase range.

More antennas usually equate to higher speeds. A wireless adapter with three antennas can have a speed of 600 Mbps. An adapter with two antennas has a speed of 300 Mbps. The router needs multiple antennas and must fully support all features of 802.11n to attain the highest speed possible.

Legacy wireless devices use Single-Input Single-Output (SISO) technology. They can only send or receive one spatial stream at a time.

Traditional Routers

Single-User MIMO Technology
Wi-Fi to one device at a time.



MU-MIMO Routers

Next-Gen Multi-User MIMO Technology
Wi-Fi to multiple devices at one, at the same speed.



Figure 30 Single and Multi-User MIMO

UDP vs TCP

Usually a video stream is somewhat fault tolerant. So if some packages get lost (due to some router along the way being overloaded, for example), then it will still be able to display the content, but with reduced quality.

If your live stream was using TCP/IP, then it would be forced to wait for those dropped packages before it could continue processing newer data. That's doubly bad:

- Old data be re-transmitted (that's probably for a frame that was already displayed and therefore worthless) and
- New data can't arrive until after old data was re-transmitted

Drawbacks of using TCP for live video:

- Typically live video-streaming appliances are not designed with TCP streaming in mind. If you use TCP, the OS must buffer the unacknowledged segments for every client. This is undesirable, particularly in the case of live events; presumably your list of simultaneous clients is long due to the singularity of the event. Pre-recorded video-casts typically don't have as much of a problem with this because viewers stagger their replay activity; therefore TCP is more appropriate for replaying a video-on-demand.

-
- IP multicast significantly reduces video bandwidth requirements for large audiences; TCP prevents the use of IP multicast, but UDP is well-suited for IP multicast.

Live video is normally a constant-bandwidth stream recorded off a camera; pre-recorded video streams come off a disk. The loss-backoff dynamics of TCP make it harder to serve live video when the source streams are at a constant bandwidth (as would happen for a live-event). If you buffer to disk off a camera, be sure you have enough buffer for unpredictable network events and variable TCP send/backoff rates. UDP gives you much more control for this application since UDP doesn't care about network transport layer drops.

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